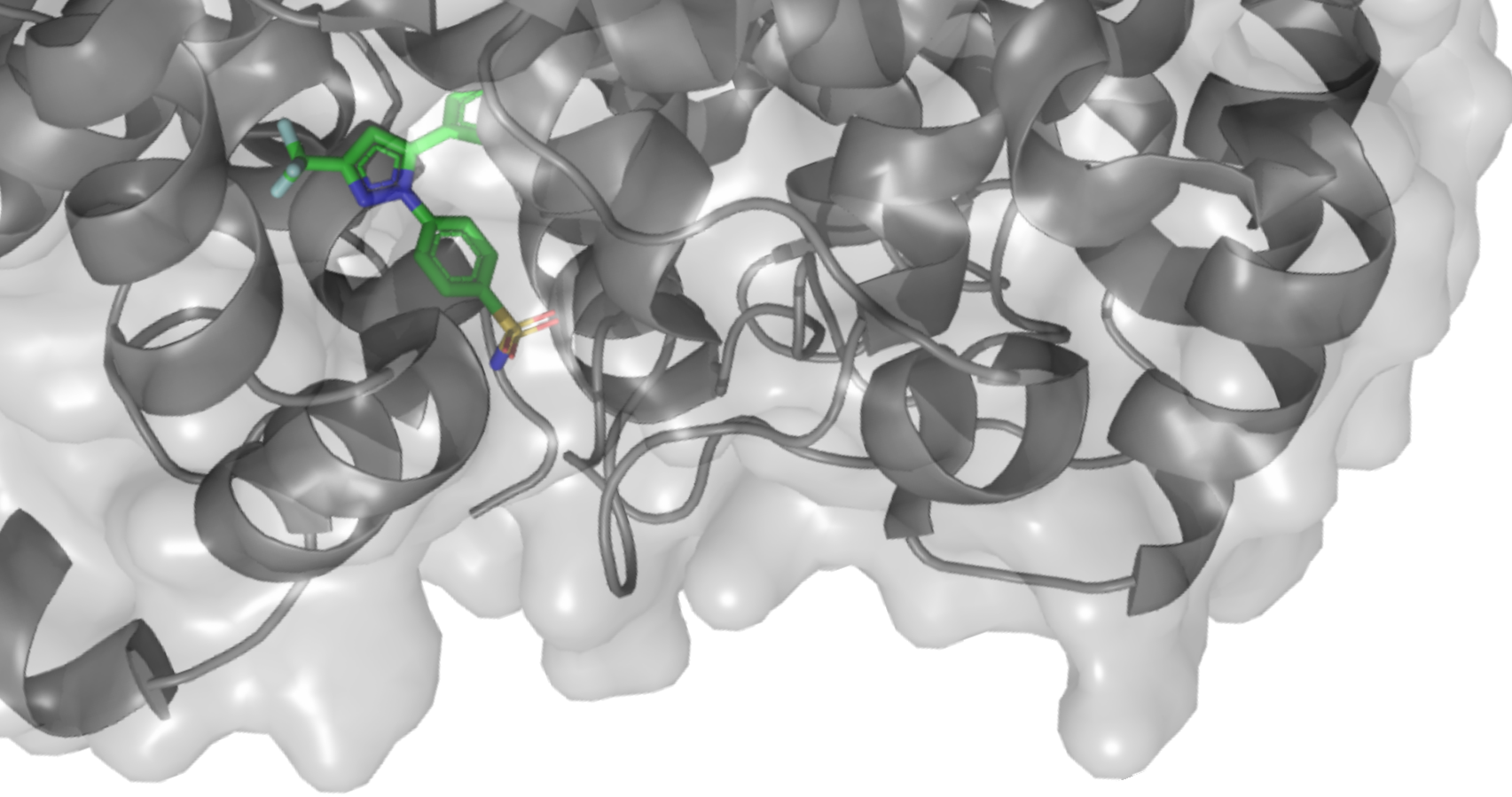


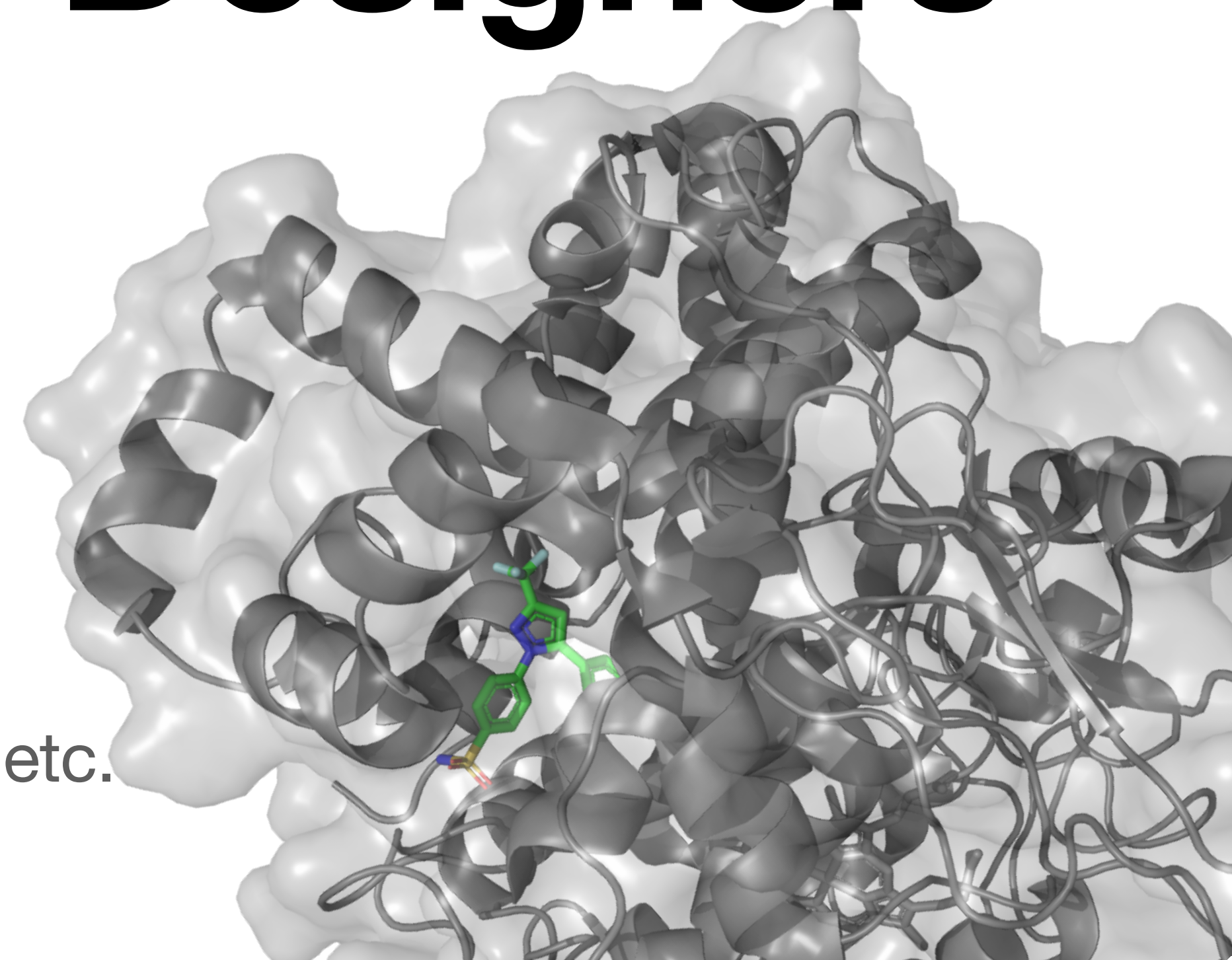


Masked Language Models are Fragment Based Drug Designers

Manas Mahale [<manas.mahale@bcp.edu.in>](mailto:manas.mahale@bcp.edu.in)
Bombay College of Pharmacy

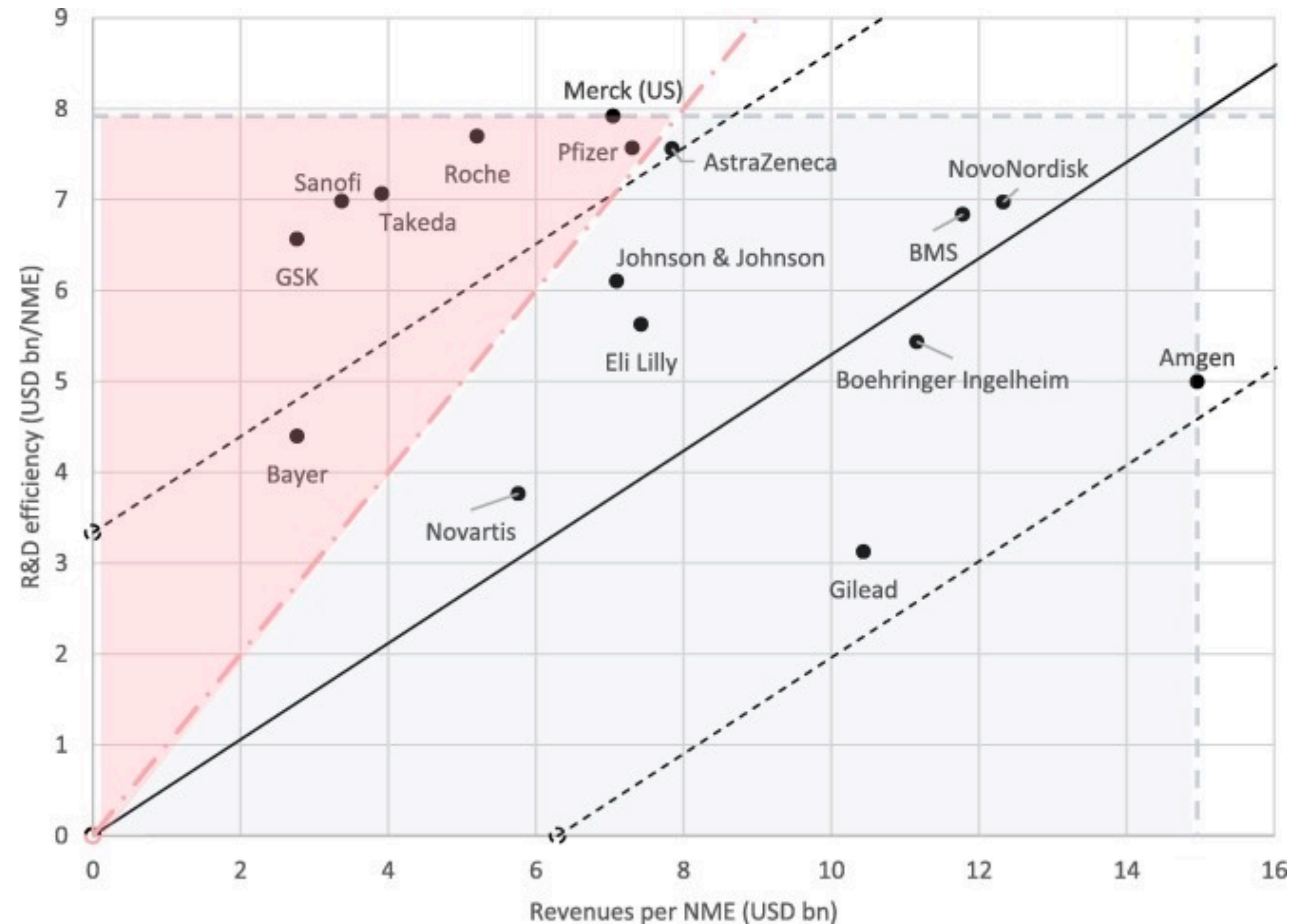
ANVESHAN National Research Convention 2023-24

Category: Health Sciences and Allied Subjects, Pharmacy, Nutrition, etc.



Context of this work

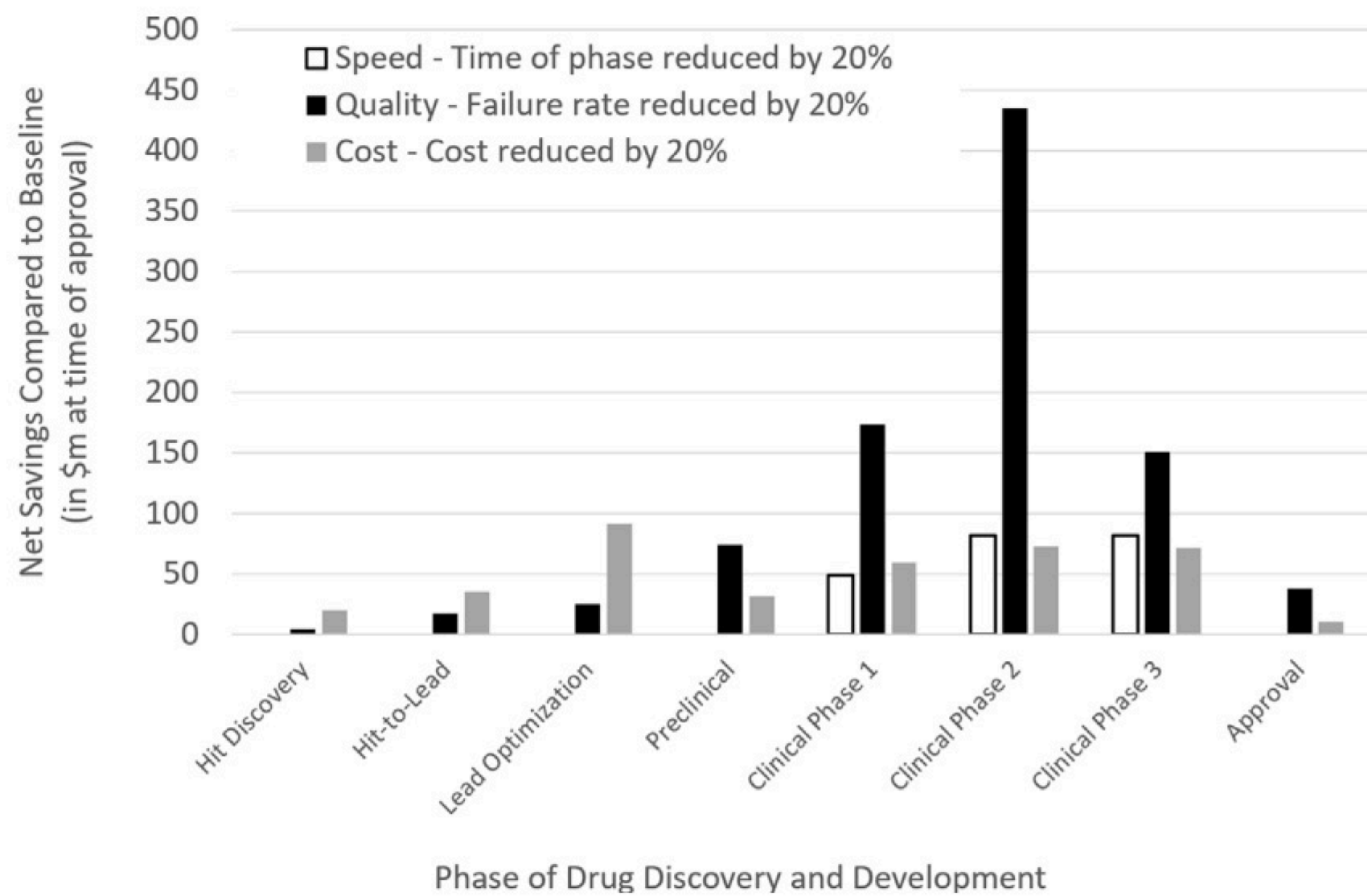
Challenges in Drug Discovery Efforts



Drug Discovery is Expensive^[*]

^[*] Schuhmacher et.al, 2023 [<https://doi.org/10.1016/j.drudis.2023.103726>]

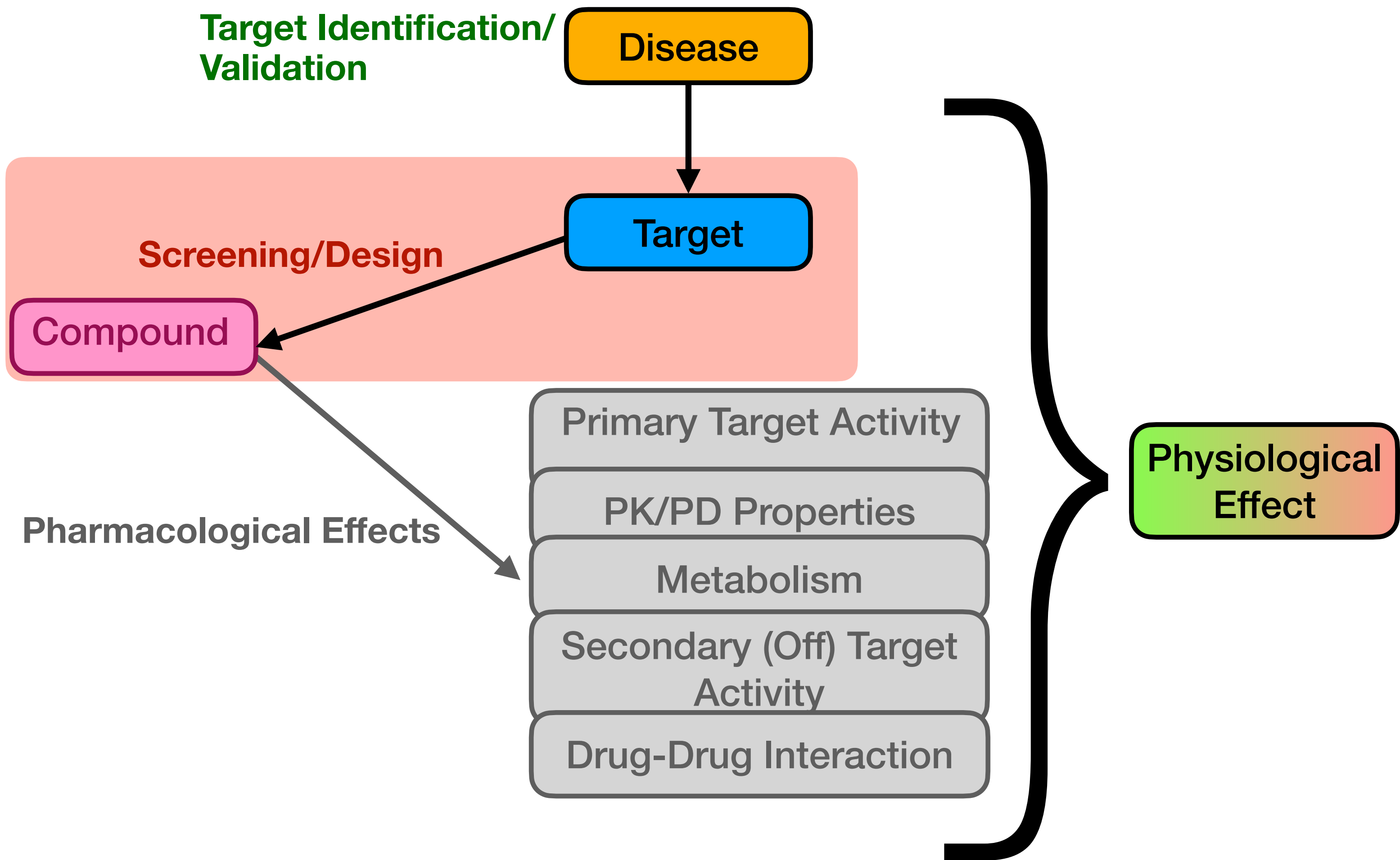
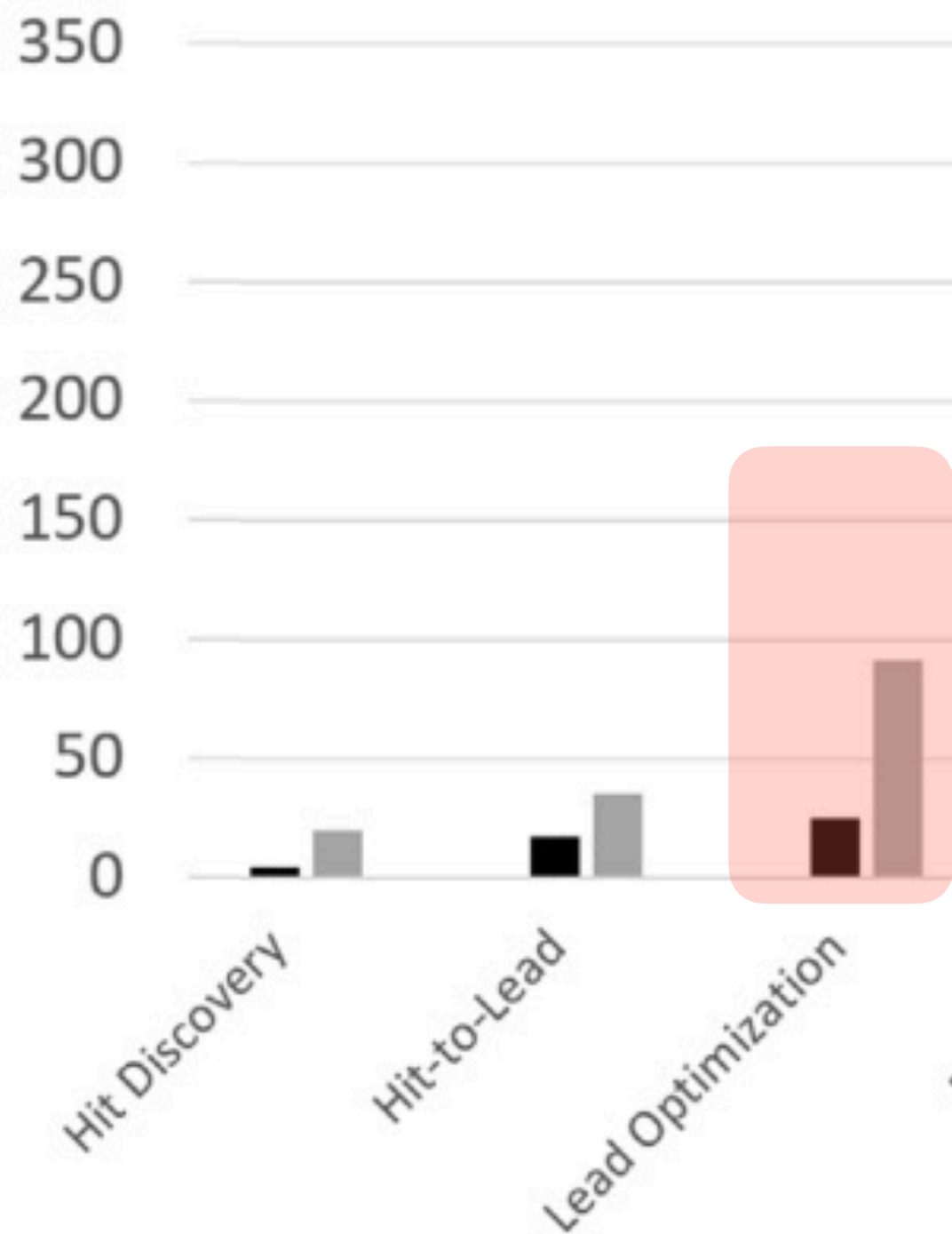
Context of this work



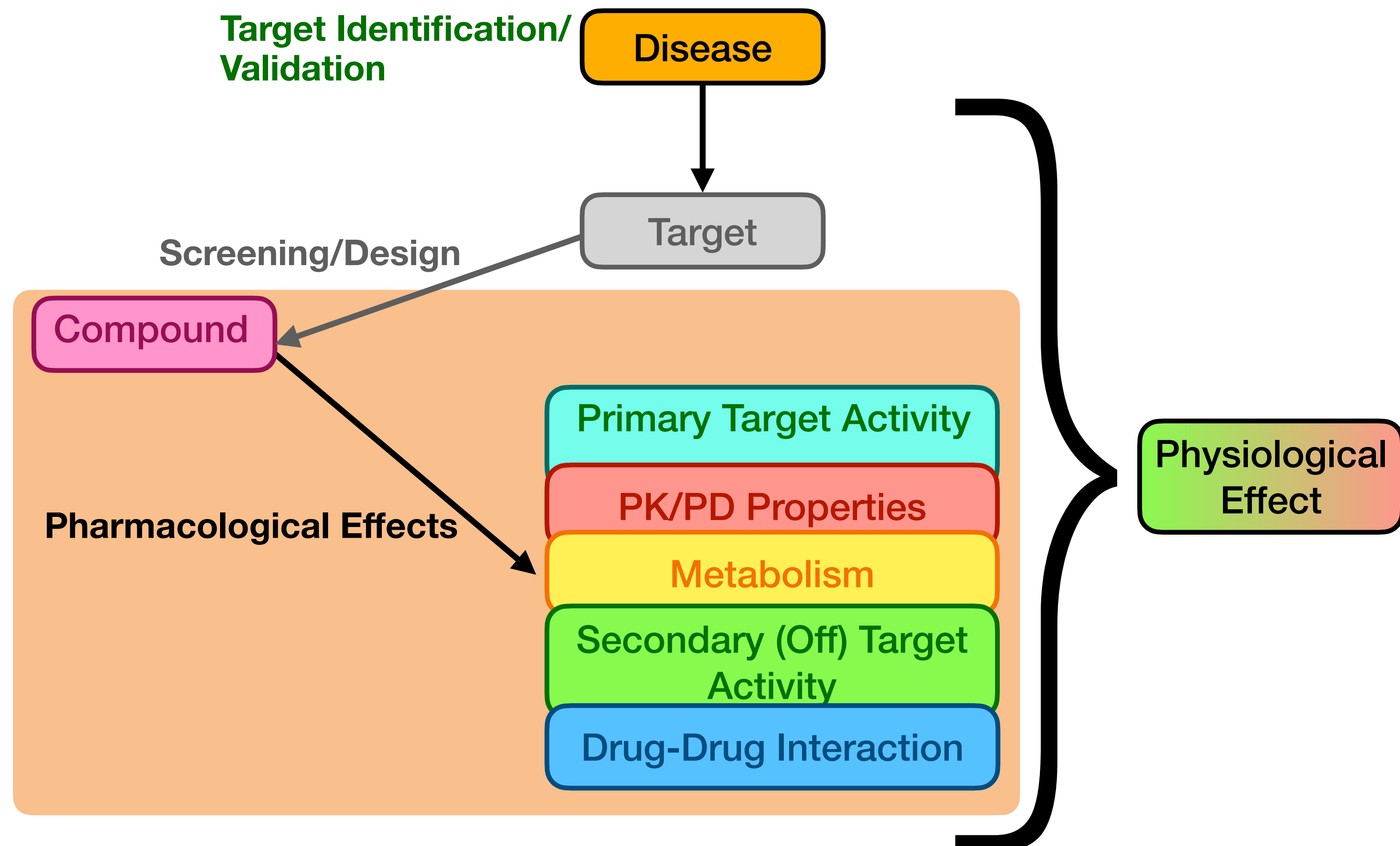
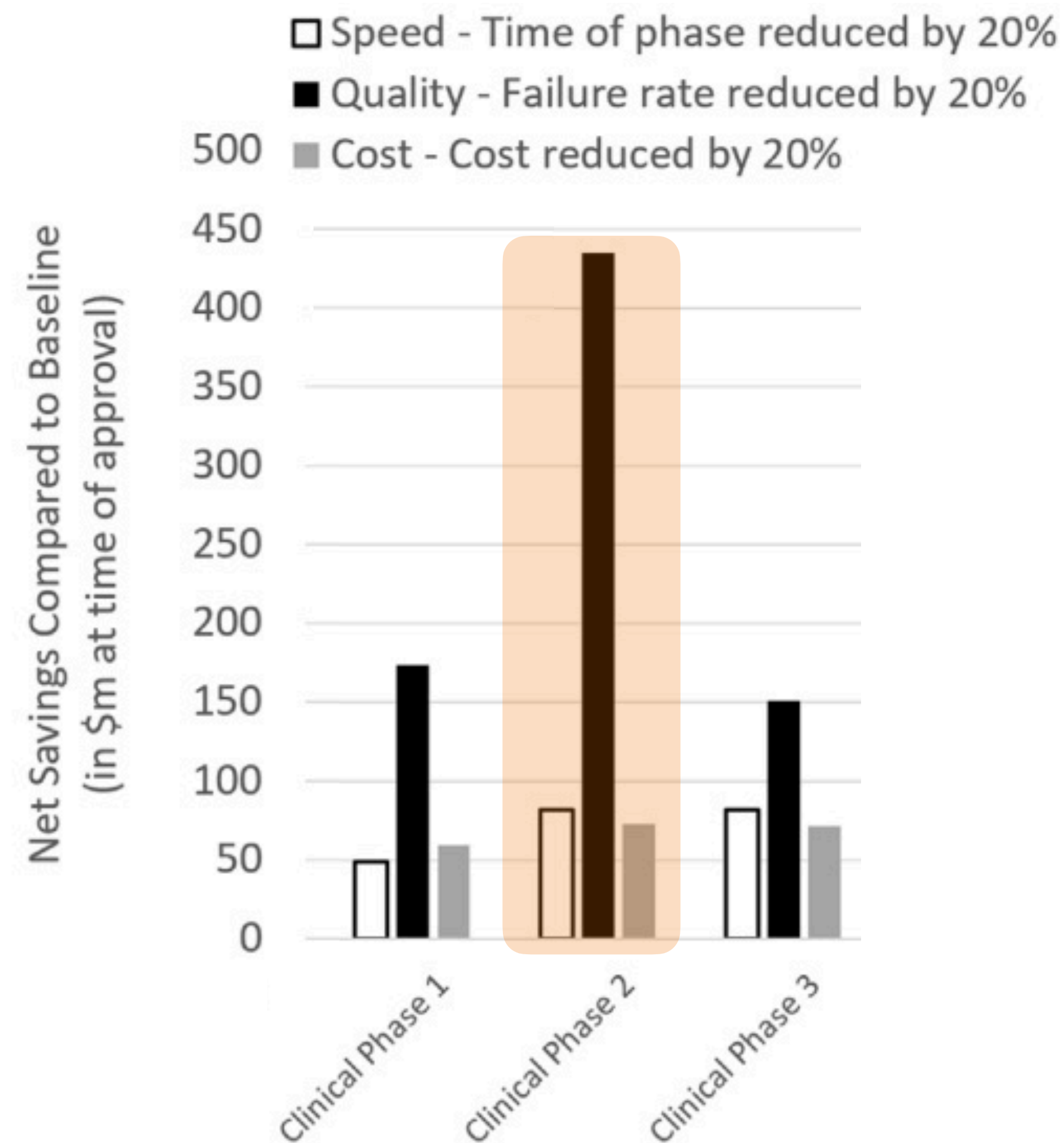
Reduction in **Cost** for Design / Screening of Lead Molecule

- Speed - Time of phase reduced by 20%
- Quality - Failure rate reduced by 20%
- Cost - Cost reduced by 20%

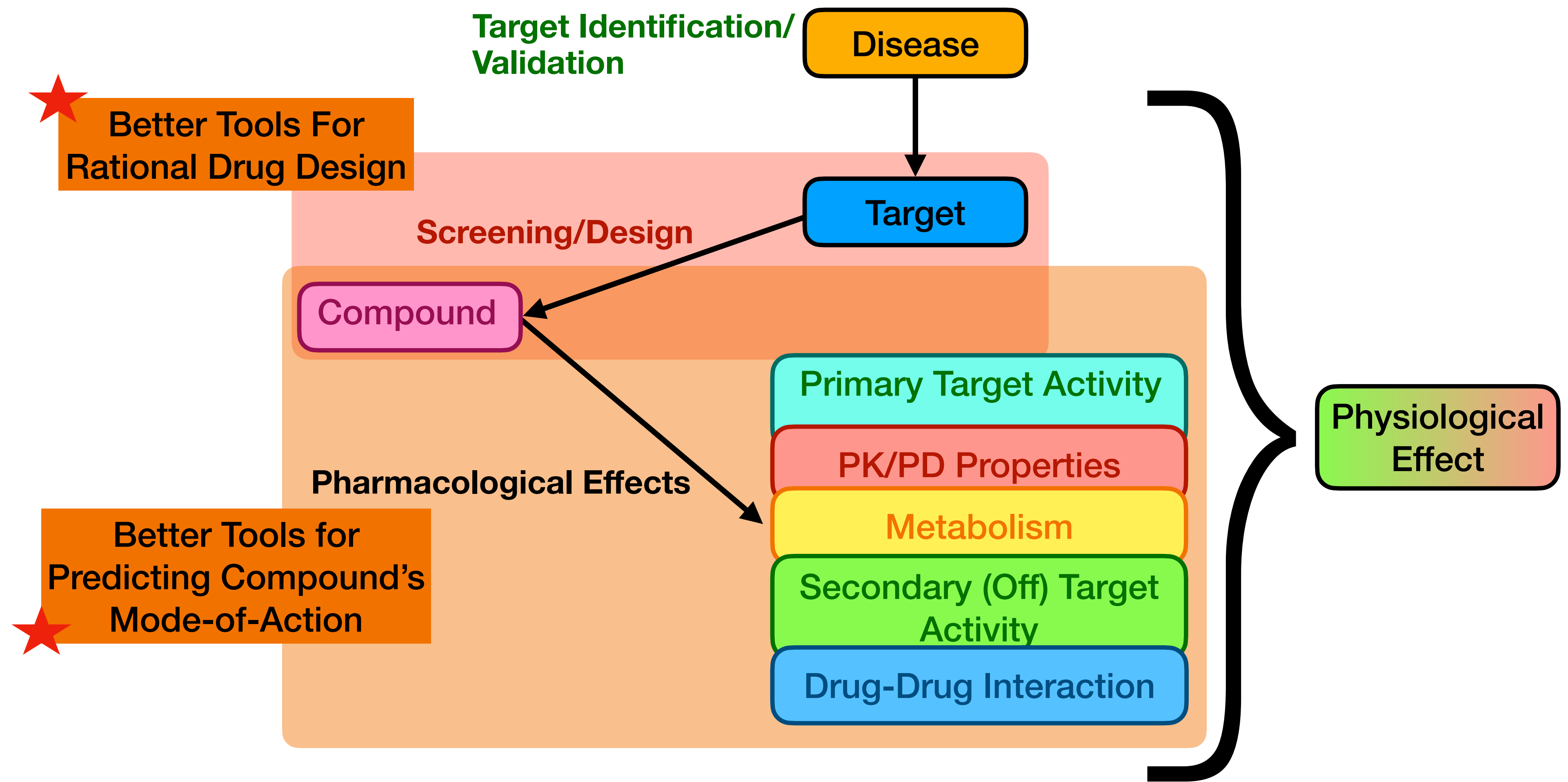
Net Savings Compared to Baseline
(in \$m at time of approval)



Improvement in *Quality* of Molecule that Enter Clinical Trials

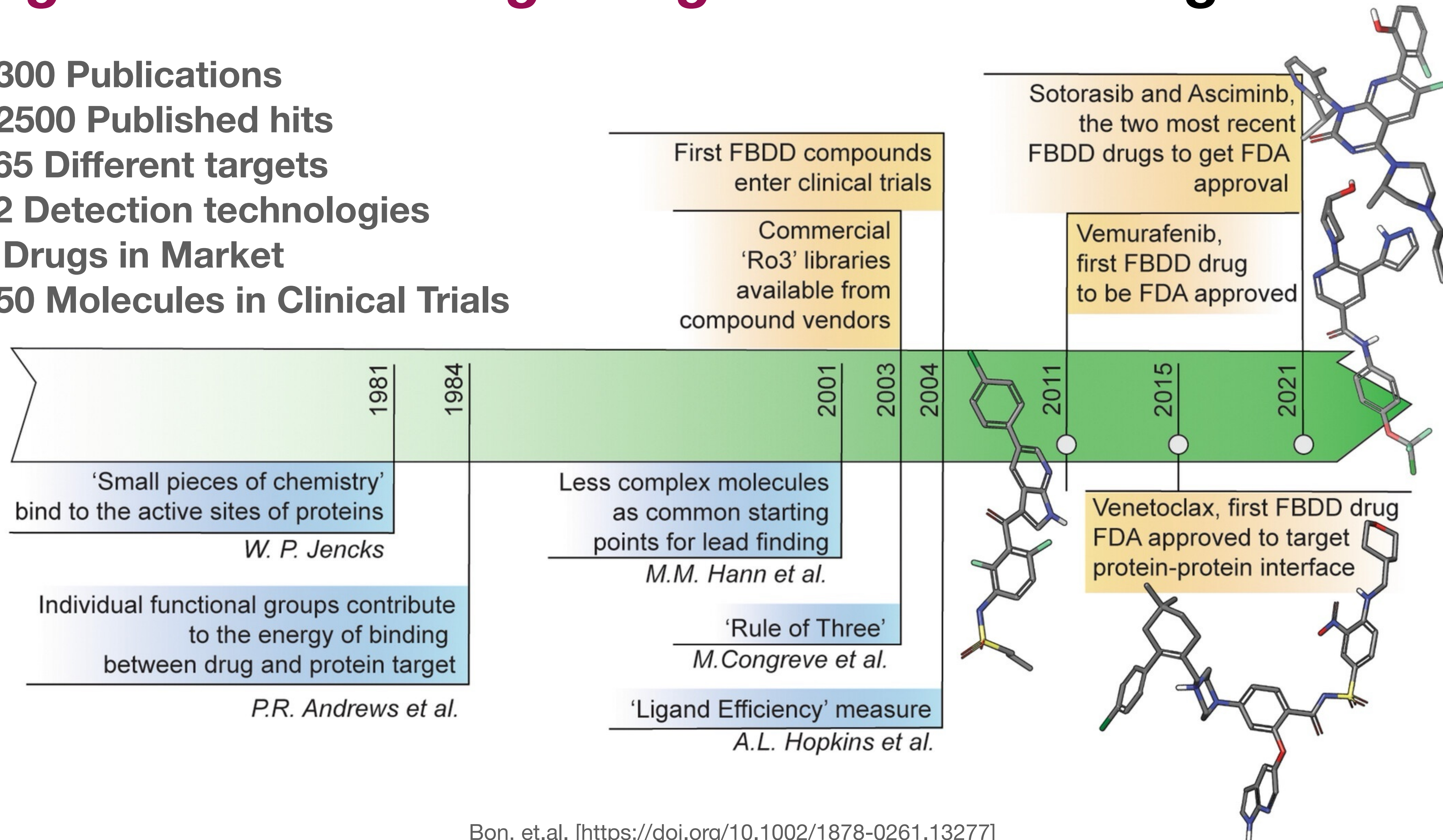


Key Opportunities in Reducing *Cost* & Improving *Quality* of Molecules



Fragment Based Drug Design as Rational Drug Discovery

- >300 Publications
- >2500 Published hits
- 265 Different targets
- 32 Detection technologies
- 6 Drugs in Market
- >50 Molecules in Clinical Trials



Fragment Based Drug Design as Rational Drug Discovery

Traditional Drug Discovery

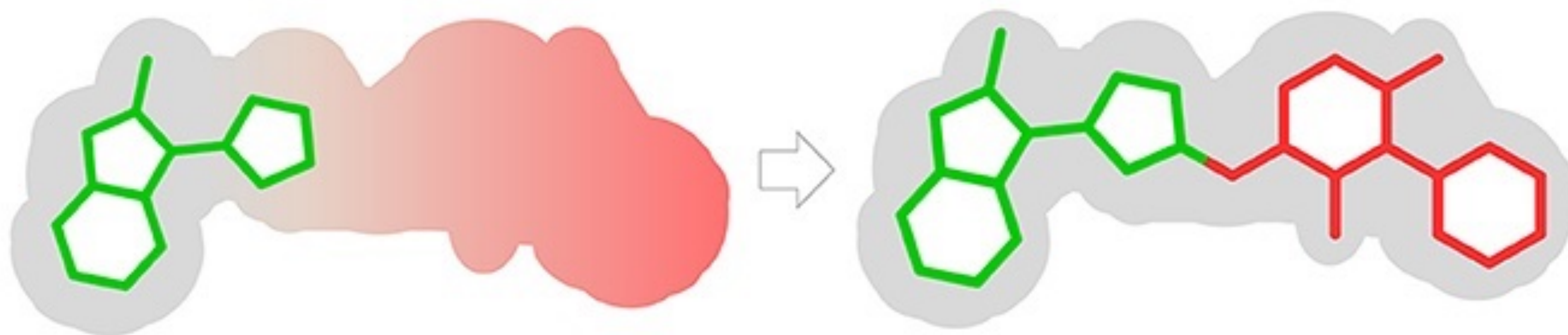


Fragment Based Drug Discovery



Fragment Based Drug Design as Rational Drug Discovery

Fragment Growth



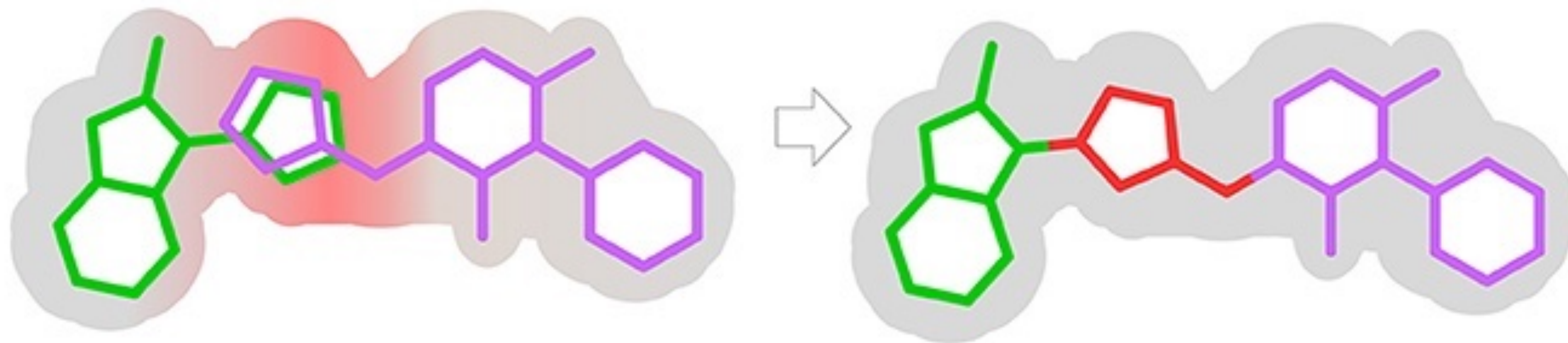
Fragment Based Drug Design as Rational Drug Discovery

Fragment Linking

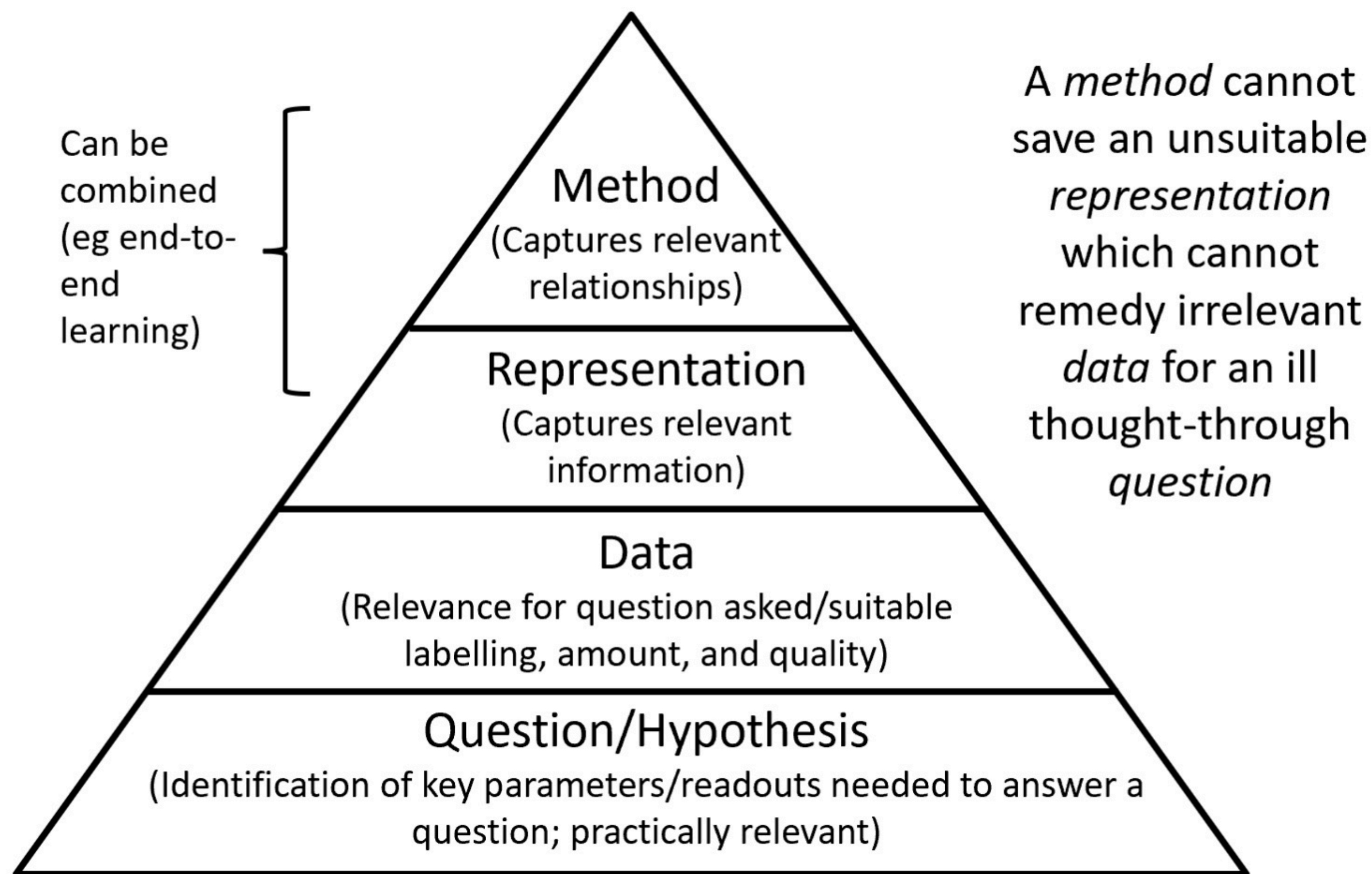


Fragment Based Drug Design as Rational Drug Discovery

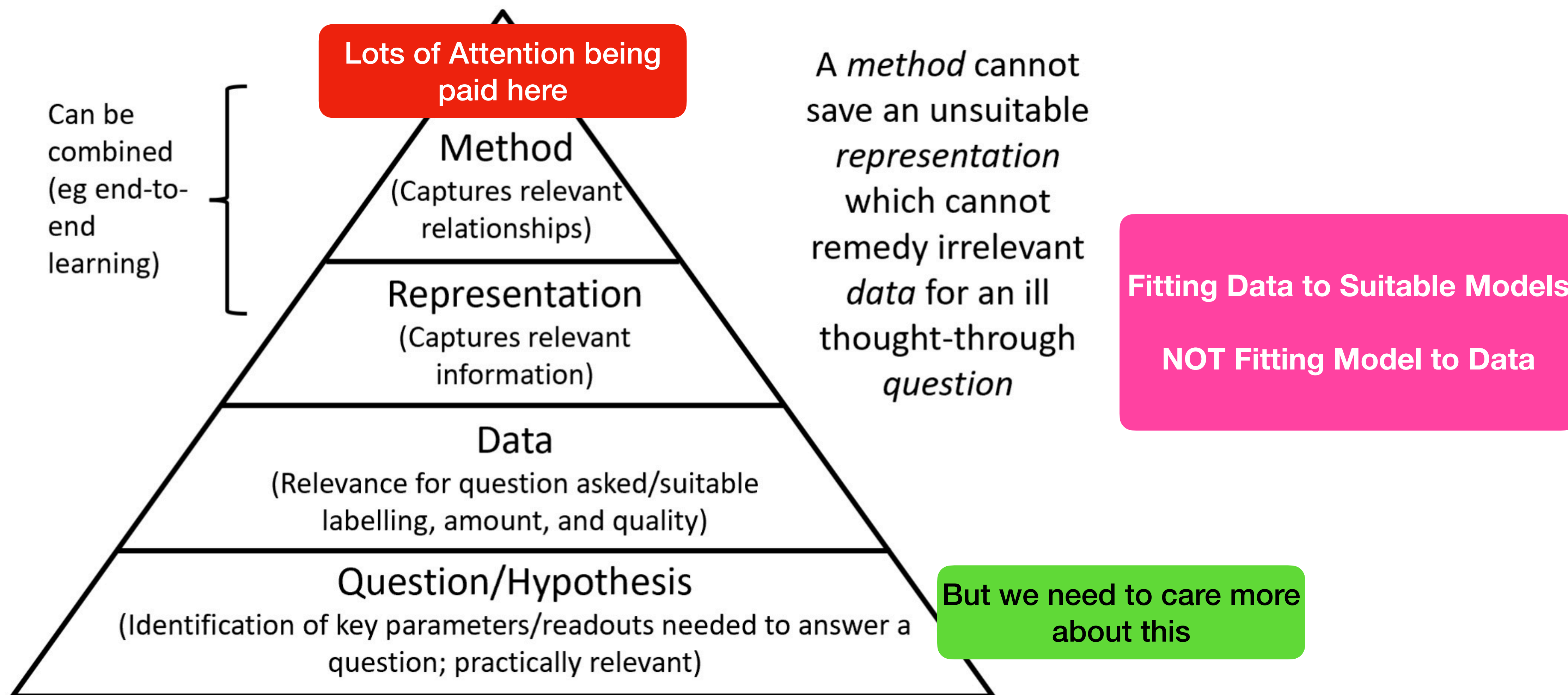
Fragment Merging



How does *AI* fit into *Fragment Based Drug Discovery* ?



How does *AI* fit into *Fragment Based Drug Discovery* ?



FragmentBERT

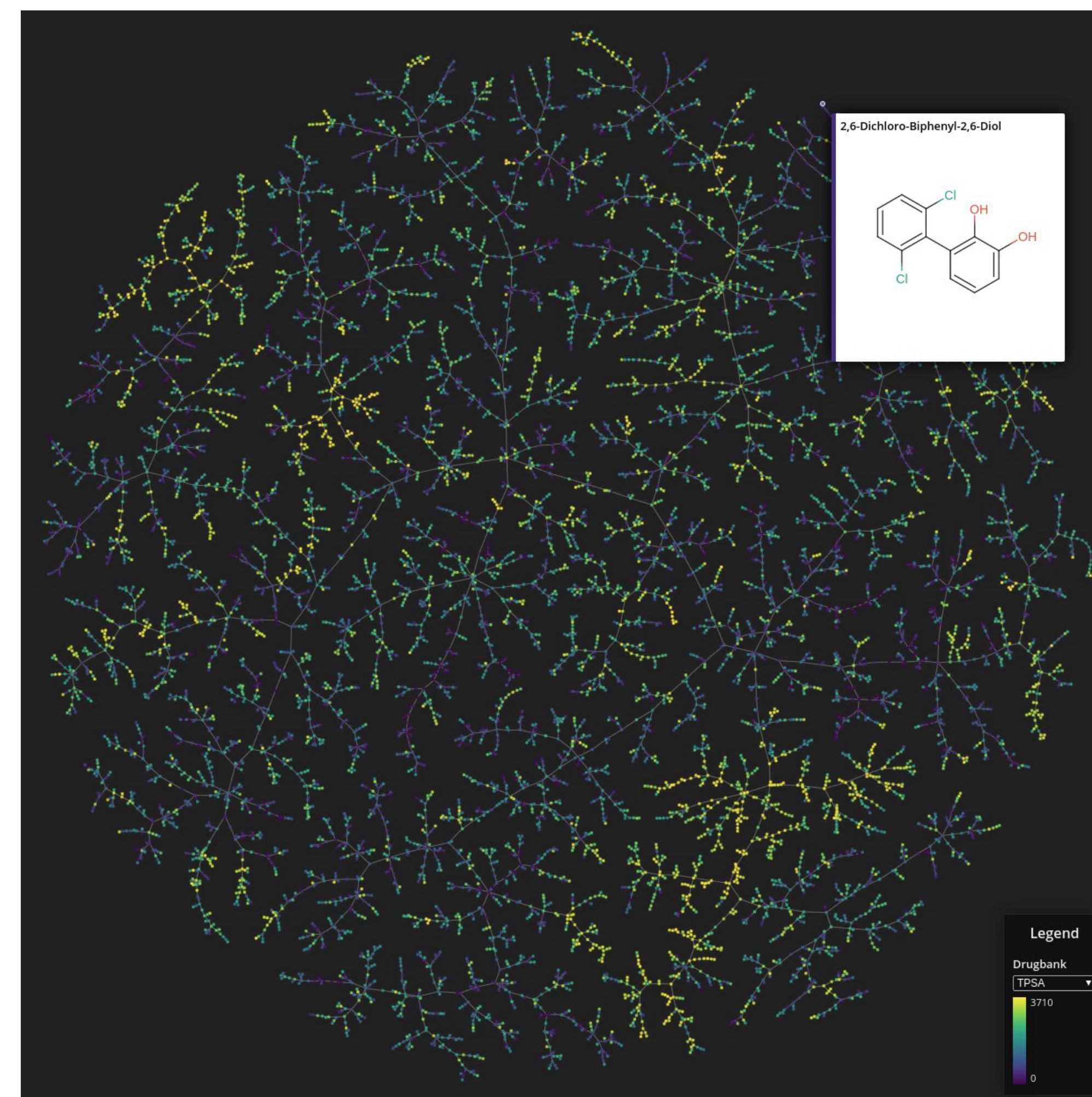
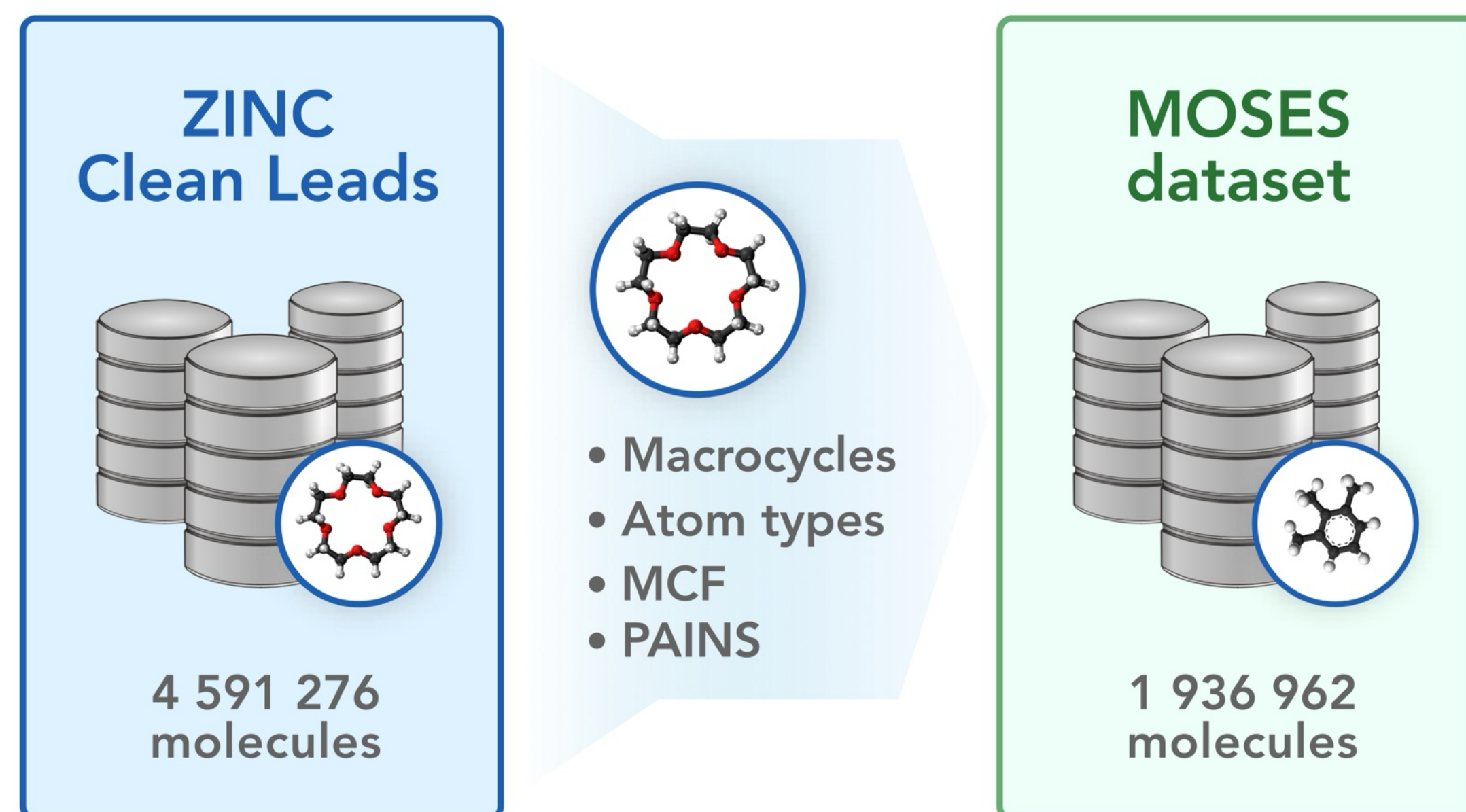
Comparison with State-of-the-Art Generative Models

Program	Algorithm	FBDD Approach	Synthetic Accessibility
AutoGrow	Docking + Genetic Algorithm	Fragment Growth	YES (Latest Version)
LUDI	Empirical Scoring	Fragment Linking	NO
LeadOP+R	Associated Reaction rules	Fragment Merging	YES
GANDI	Genetic Algorithm	Fragment Linking	NO
LigBuilder 2	Genetic Algorithm	Fragment Linking & Fragment Growth	YES**
ADAPT	Genetic Algorithm	Fragment Growth	NO
FragmentBERT (This Work)	<i>Masked Language Model</i>	Fragment Linking, Fragment Growth, Fragment Merging	YES+

** Synthetic Accessibility Model Based (SAScore, etc.)

+BRICS Based

Generating Synthetically Accessible Molecules by Training on *Drug-Like Chemical Space*

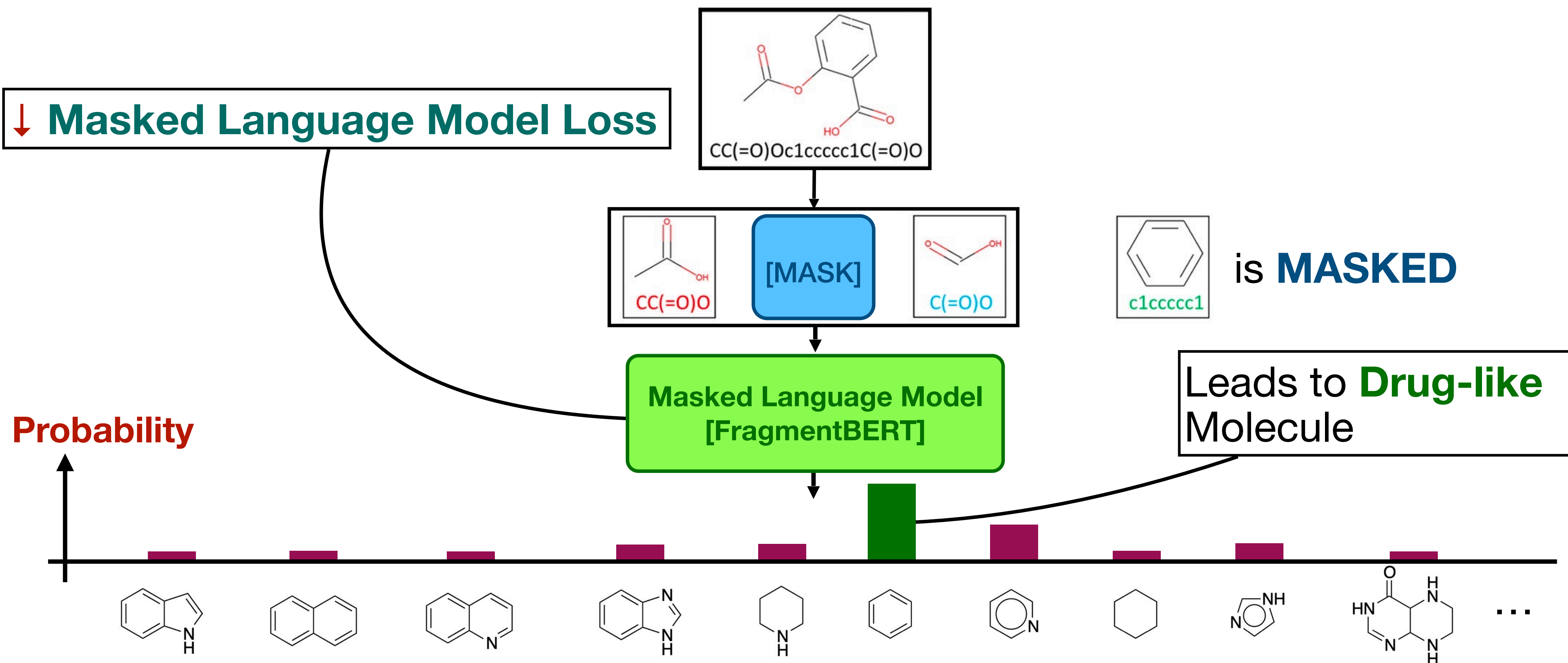


Drugbank tMAP^[*] Visualization

[*] <https://tmap.gdb.tools>

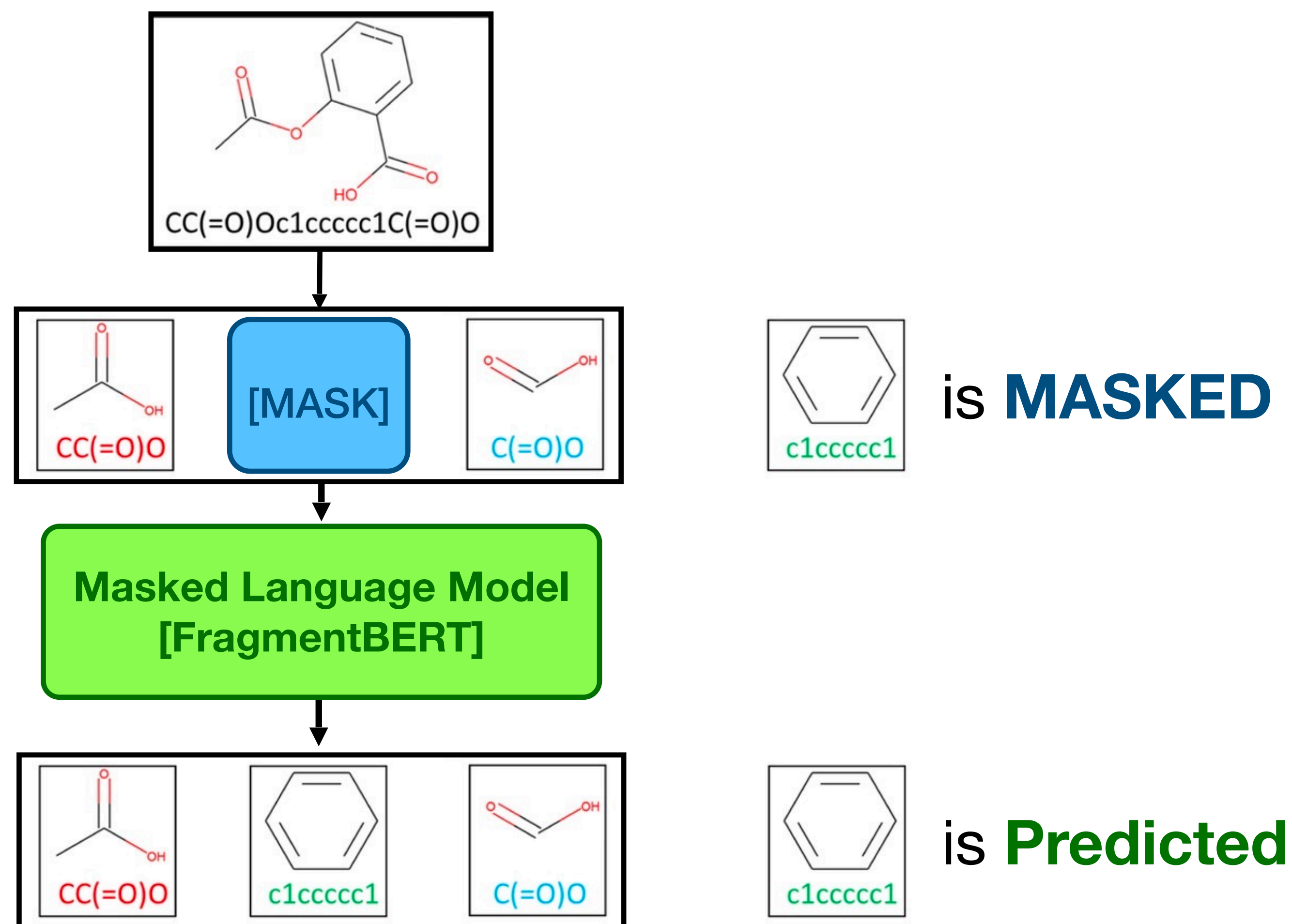
FragmentBERT

Learning to Predict Correct Masked Fragment Generates Drug-like Molecules !!

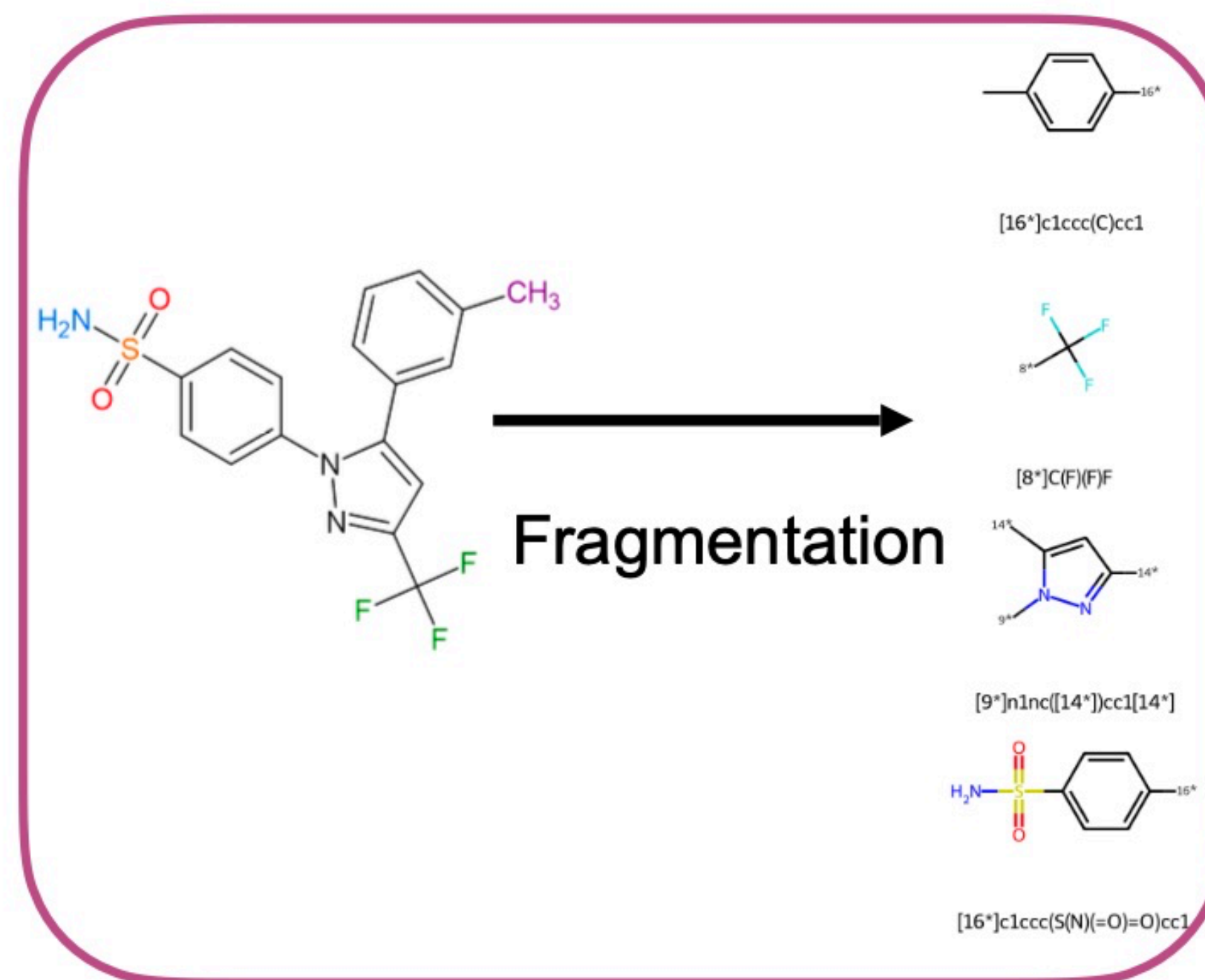
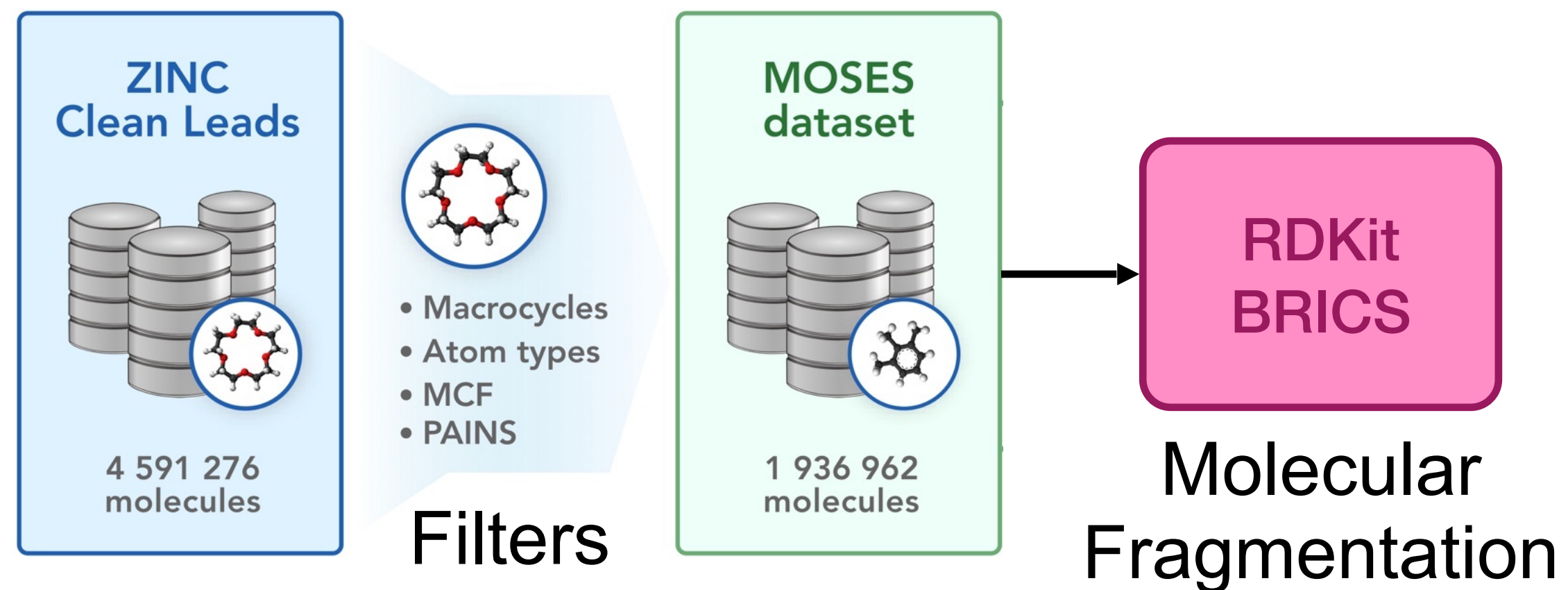


FragmentBERT

Learning to Predict Correct Masked Fragment Generates Drug-like Molecules !!



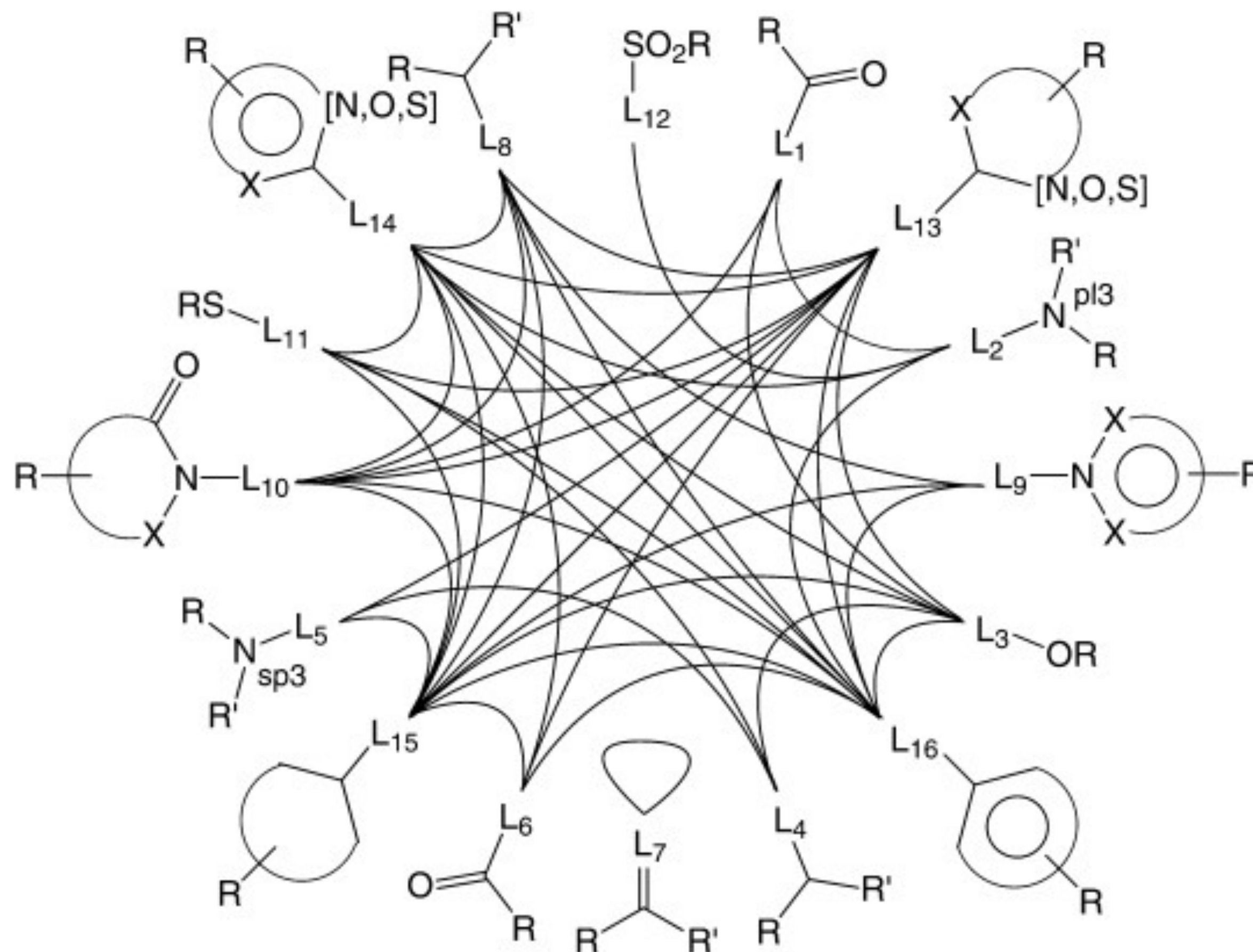
FragmentBERT: *Model Design*



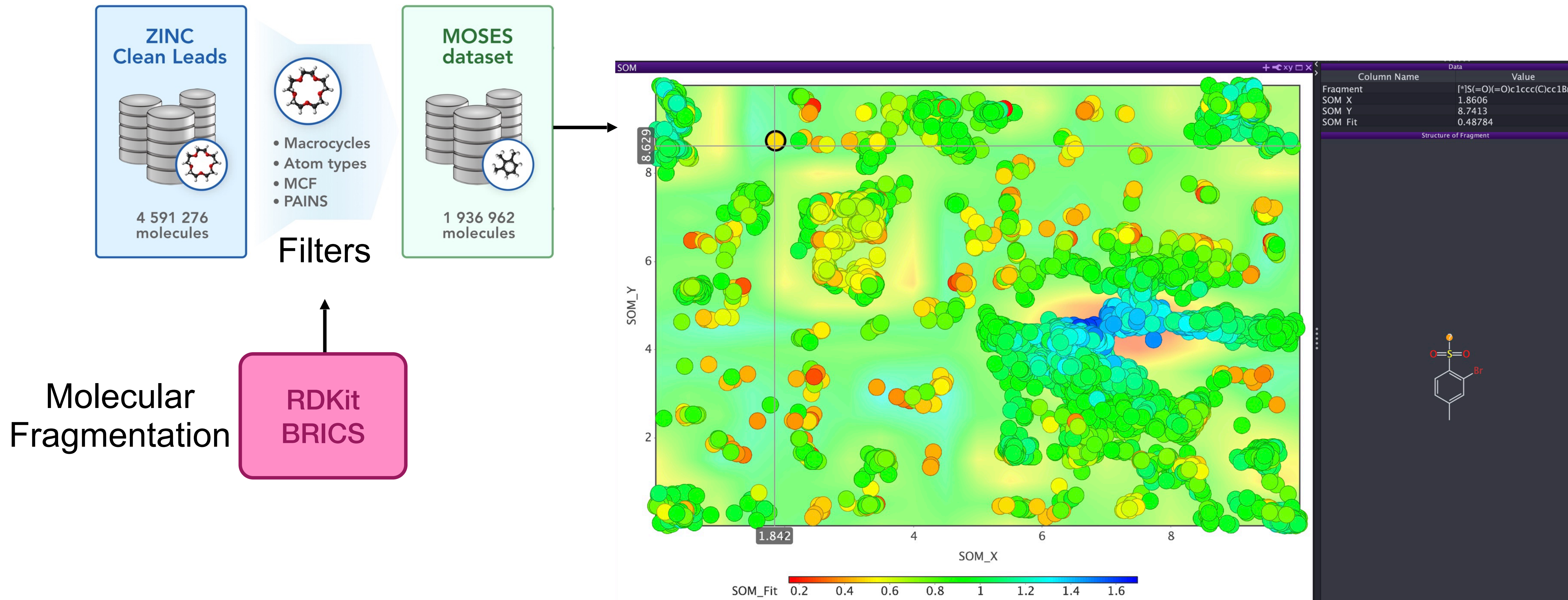
FragmentBERT: *Model Design*

RDKit
BRICS

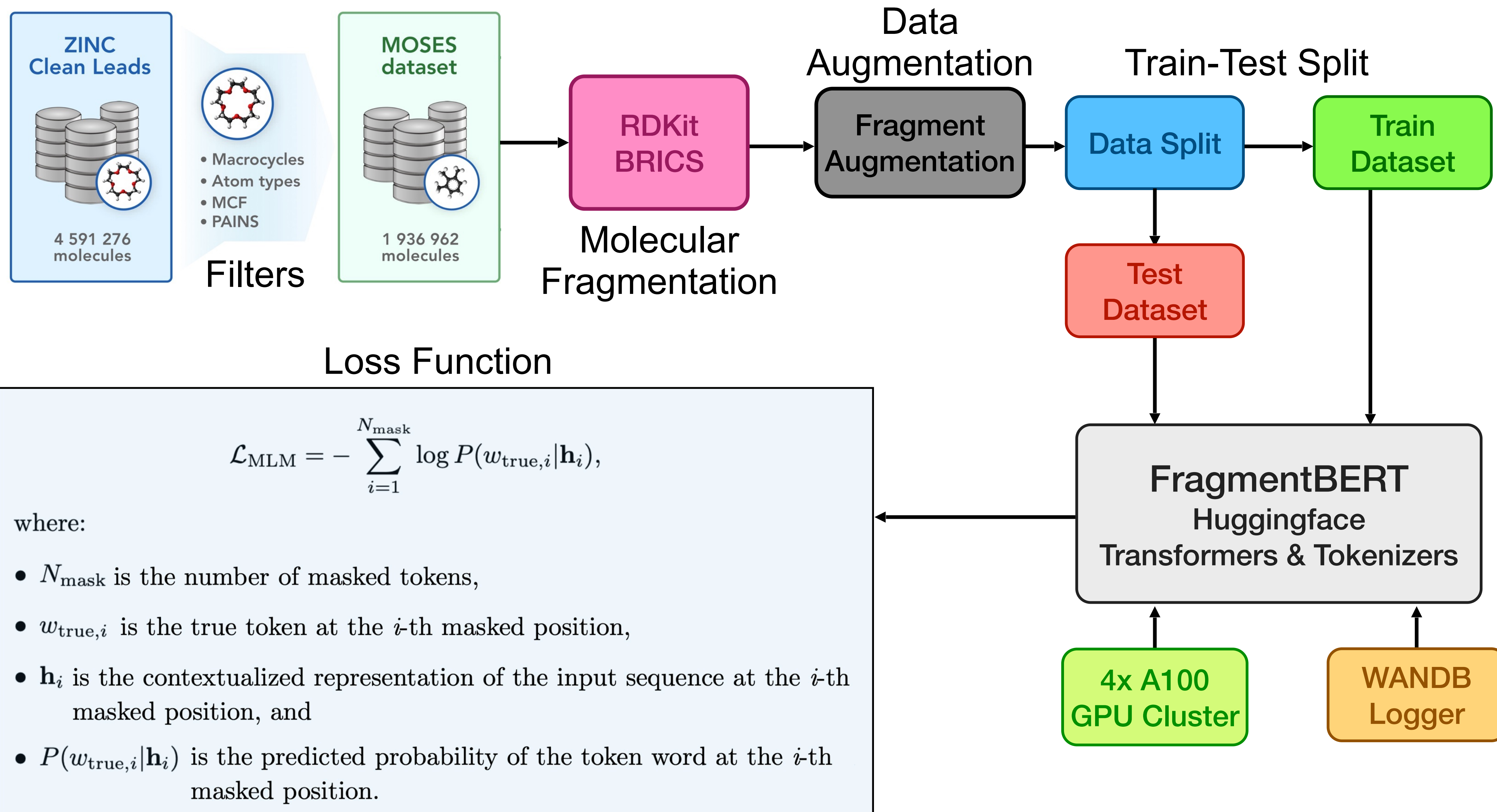
Molecular
Fragmentation



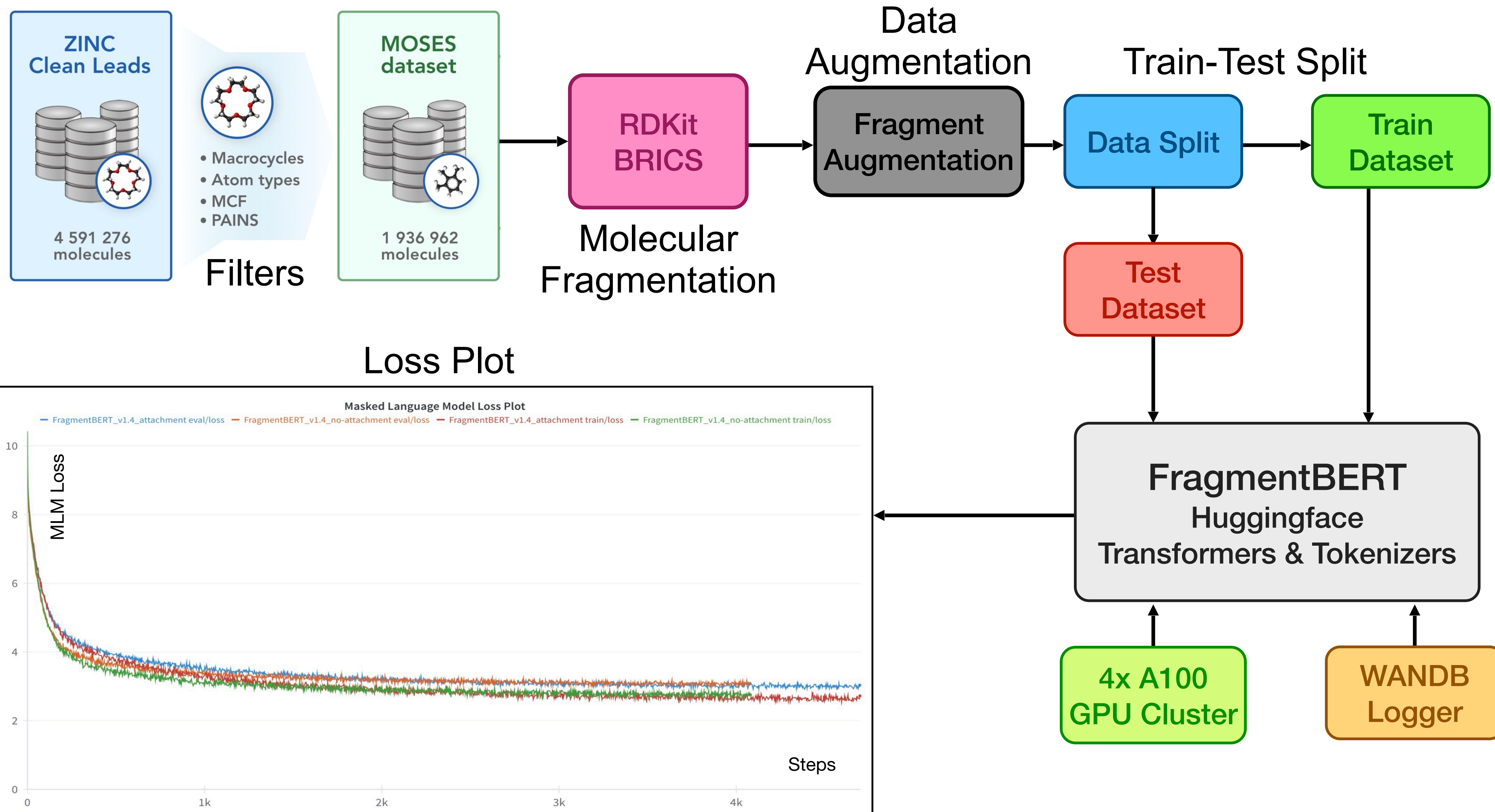
FragmentBERT: *Model Design*



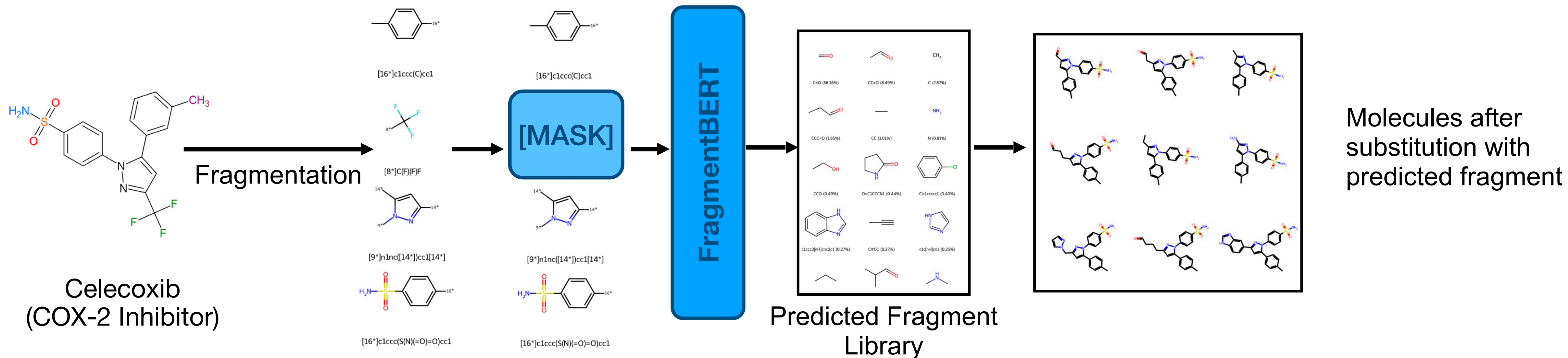
FragmentBERT: *Model Design*



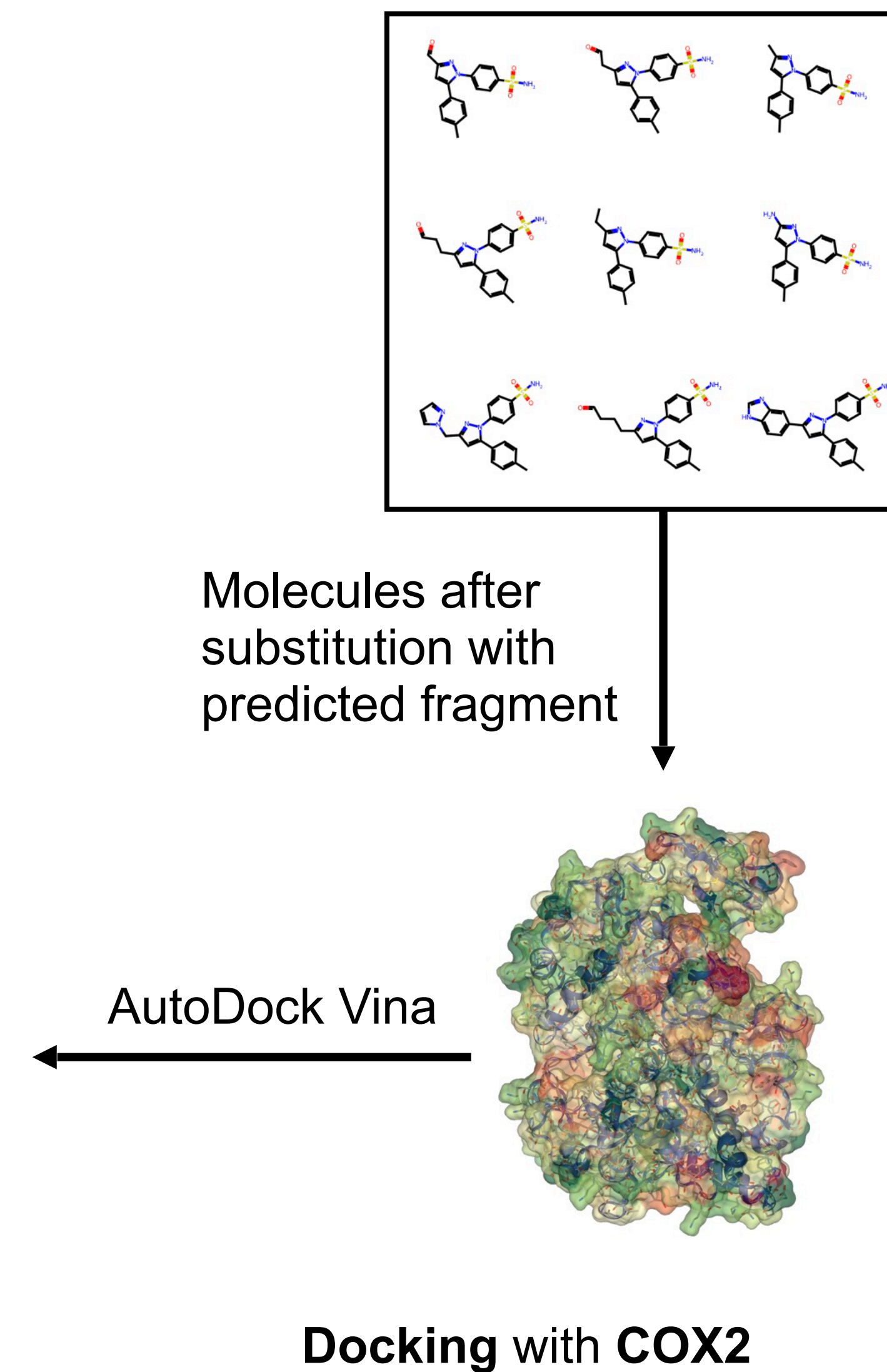
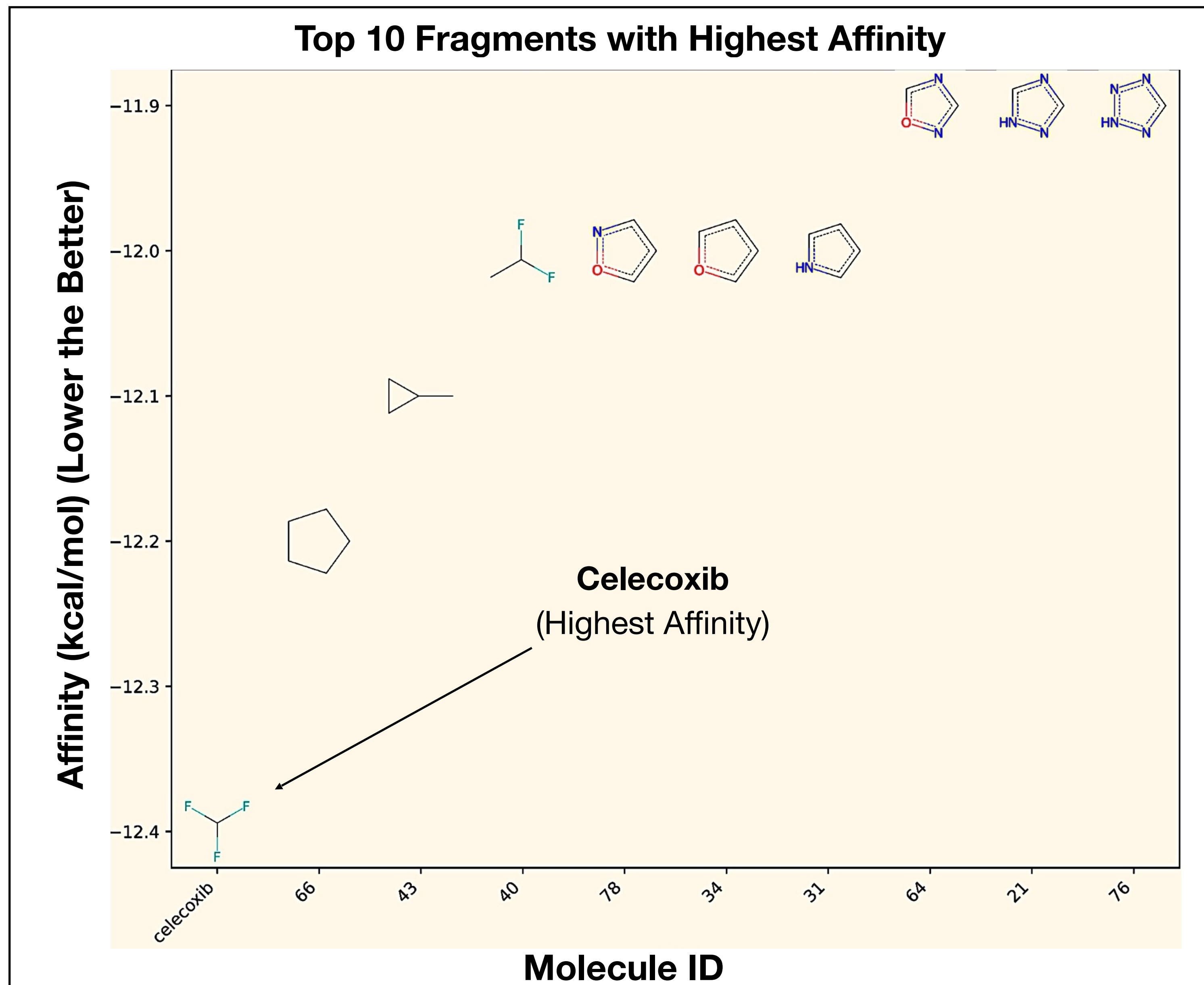
FragmentBERT: *Model Design*



FragmentBERT: *Fragment Masking*

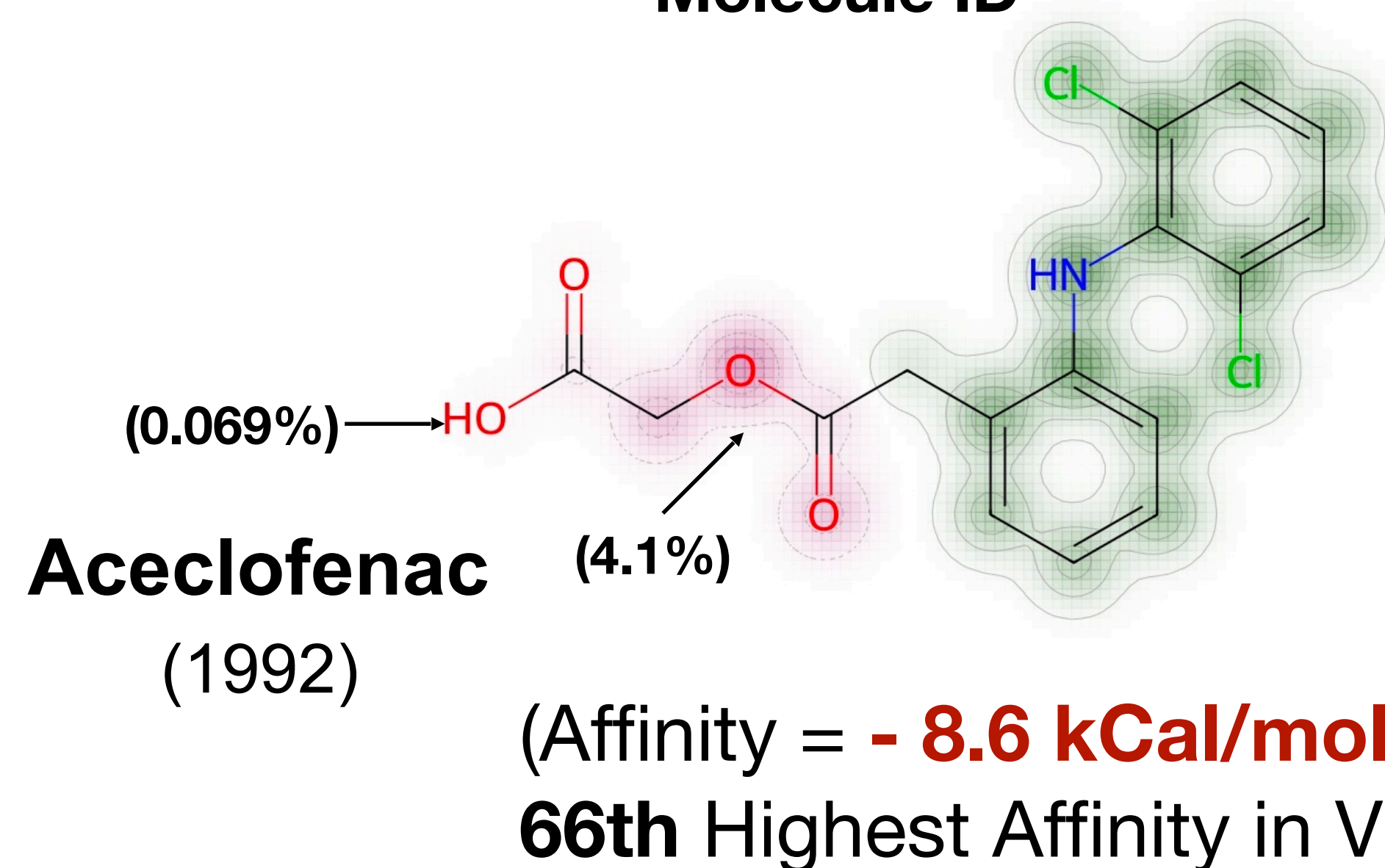
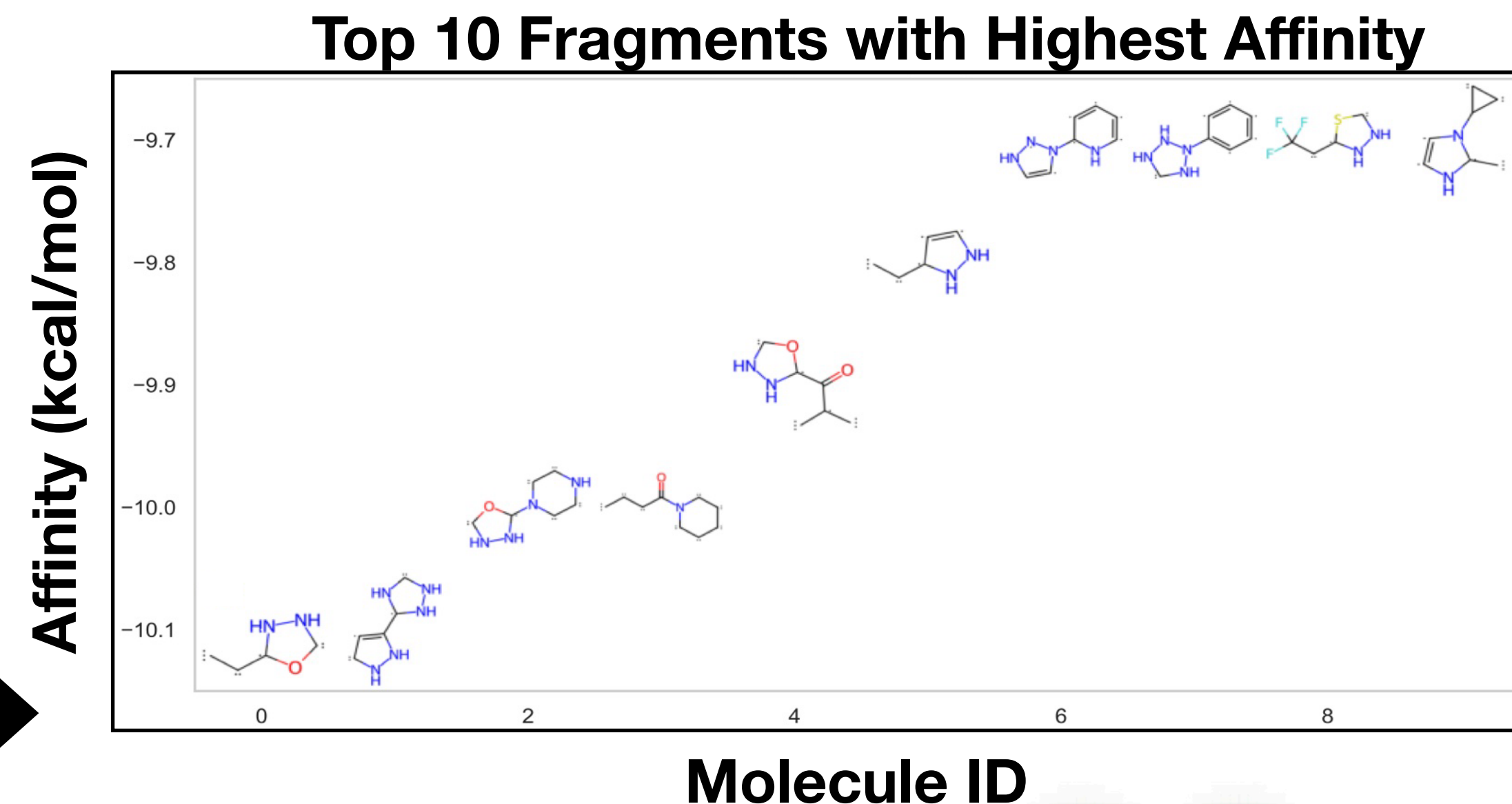
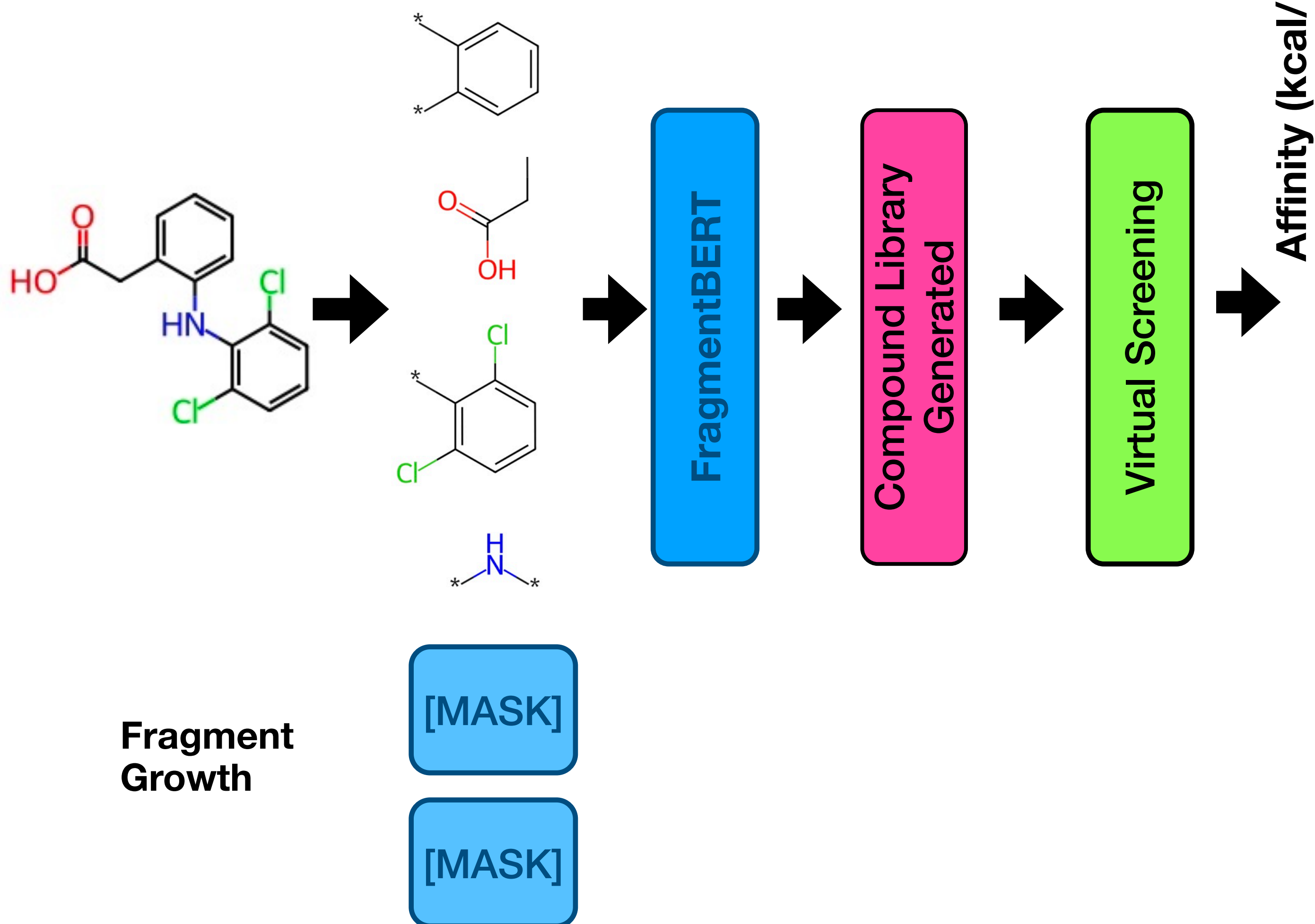


FragmentBERT: *Fragment Masking*



Retrospective Validation: *Fragment Growth*

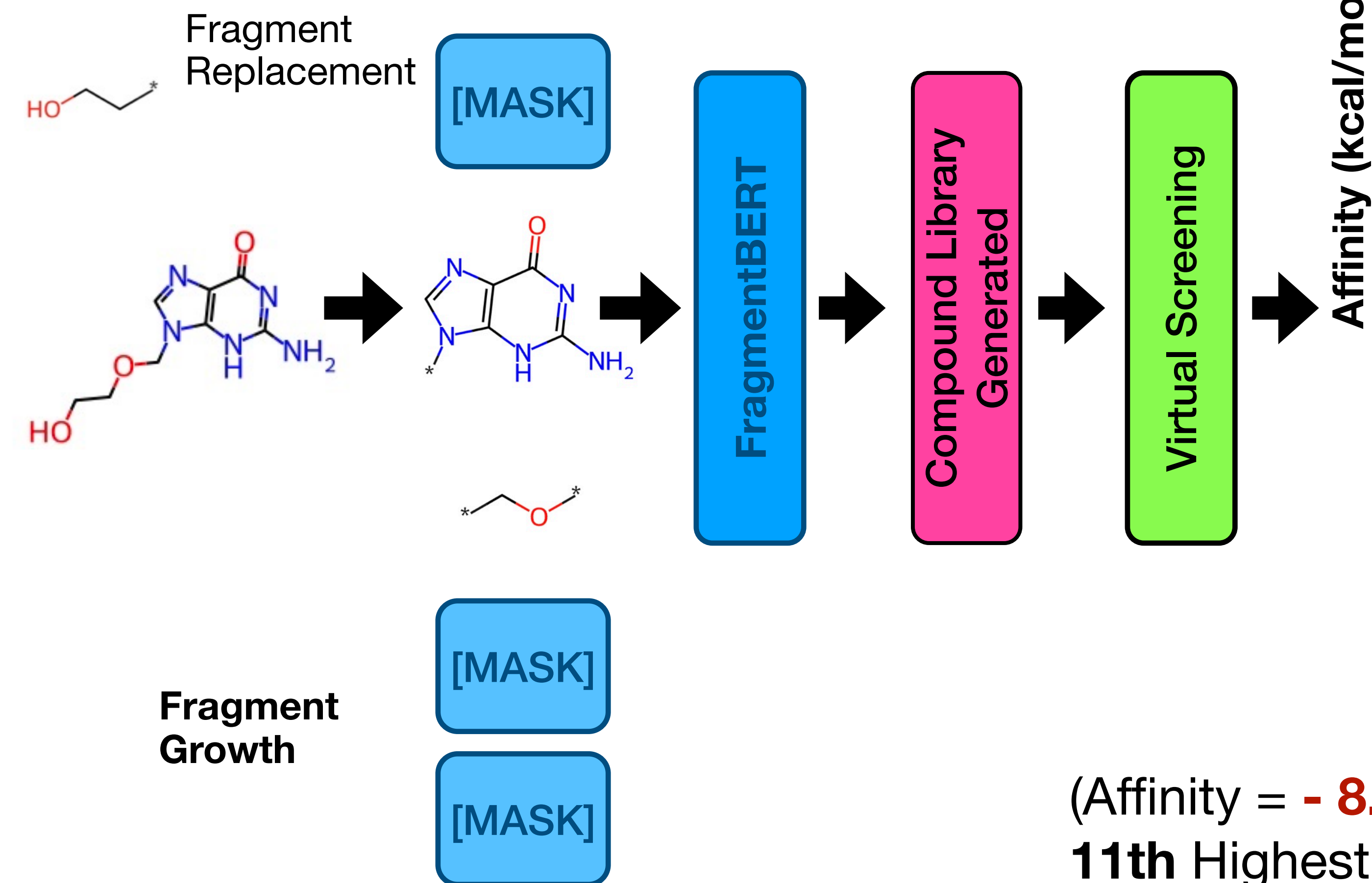
Diclofenac (Affinity = **- 8.4 kCal/mol**)
(1988) **74th** Highest Affinity in VS



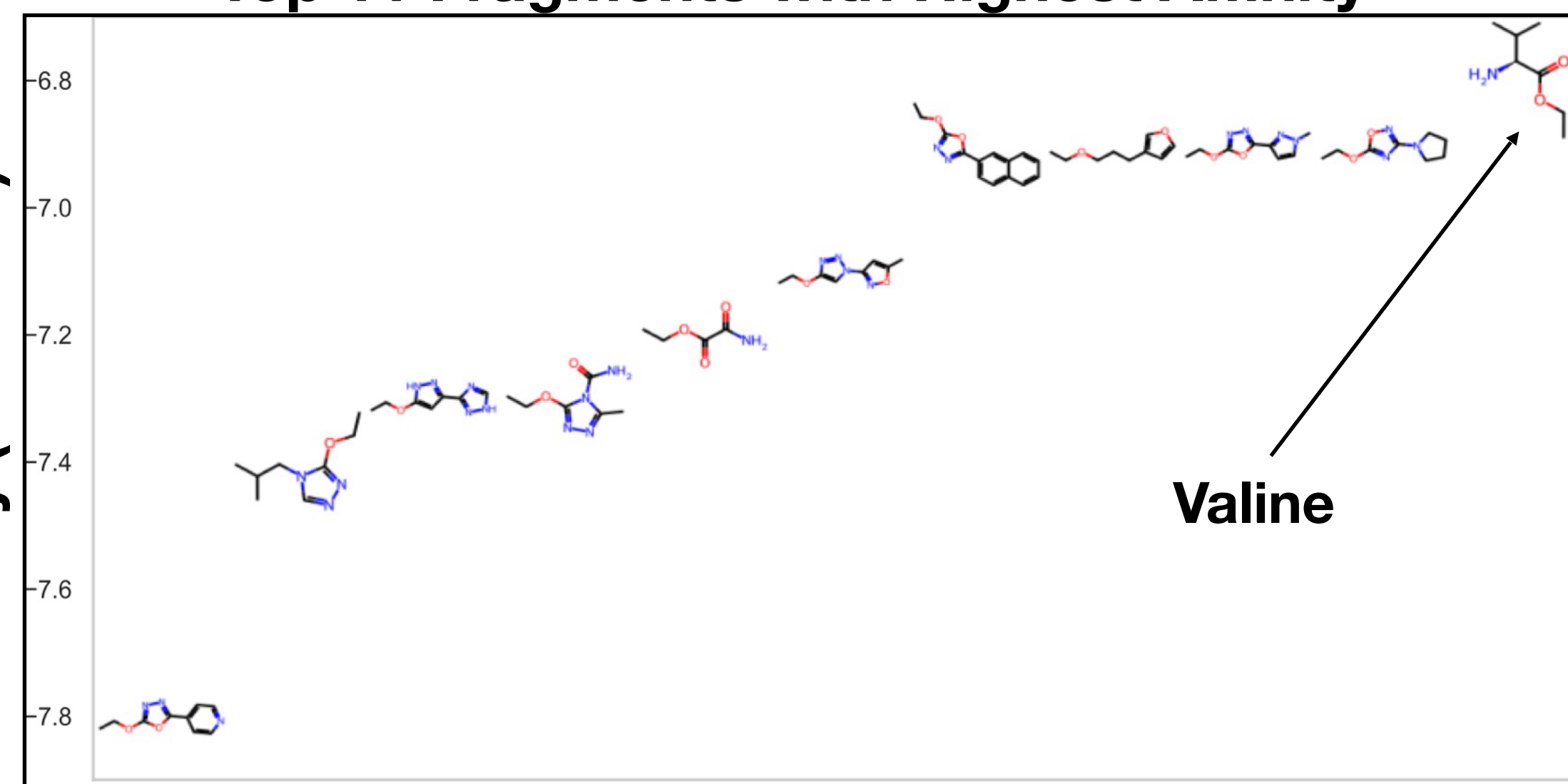
VS = Virtual Screening

Retrospective Validation: *Fragment Growth* + *Fragment Replacement*

Acyclovir (Affinity = **- 6.8 kCal/mol**)
(1974) **28th** Highest Affinity in VS

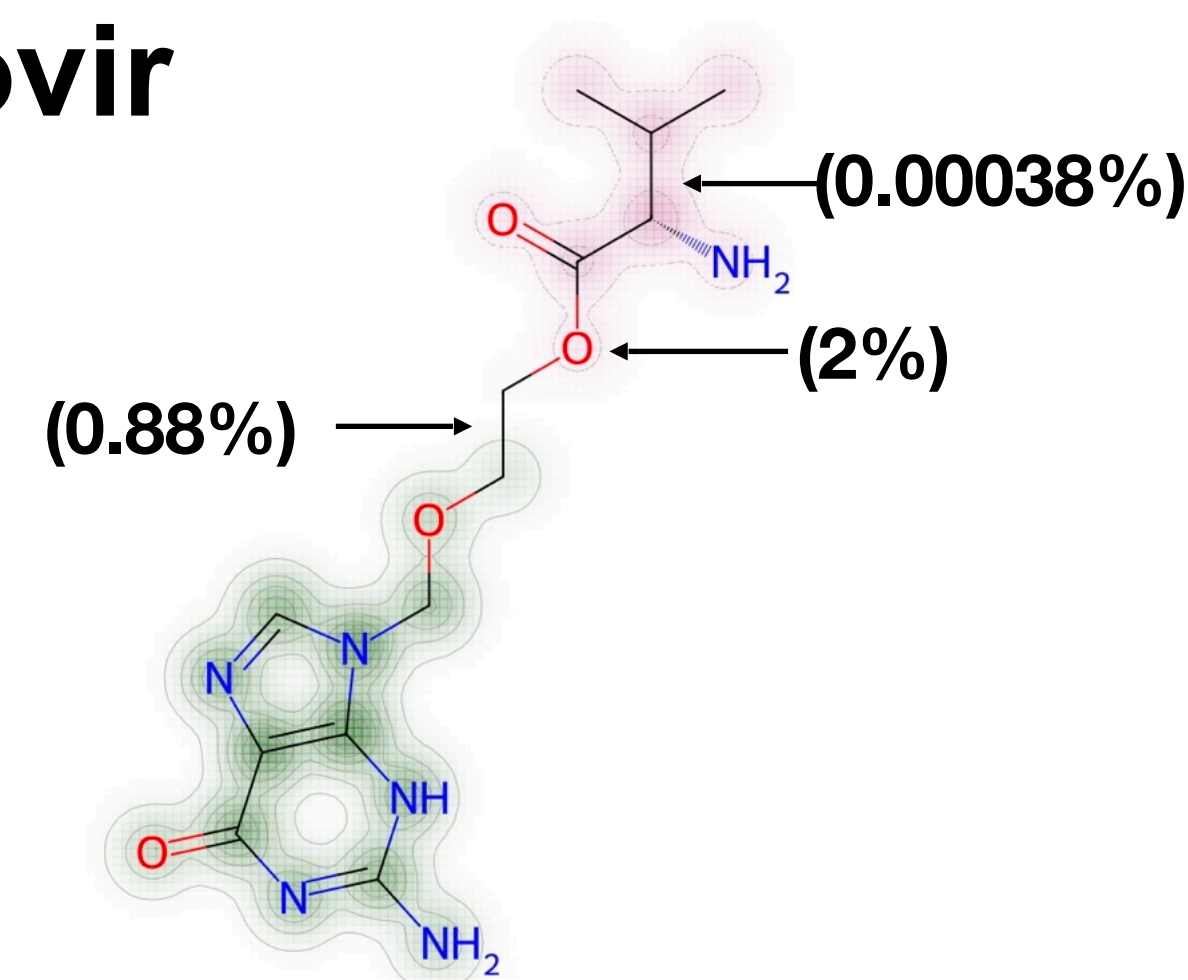


Top 11 Fragments with Highest Affinity



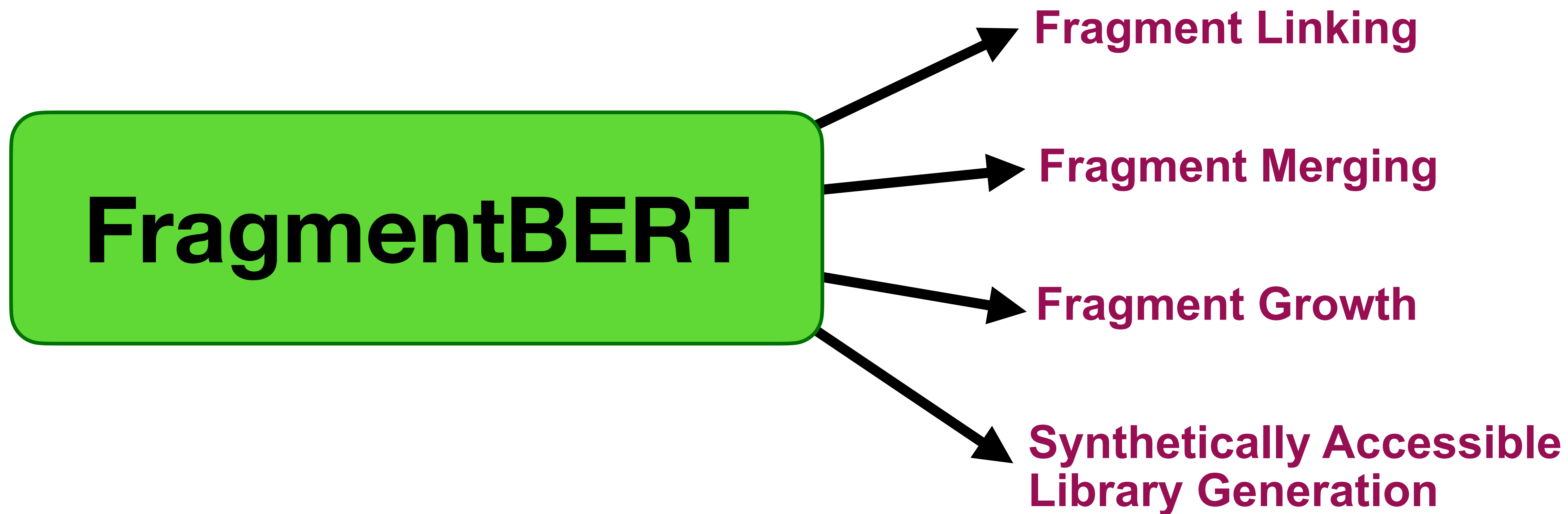
Valaciclovir
(1995)

(Affinity = **- 8.6 kCal/mol**)
11th Highest Affinity in
Virtual Screening

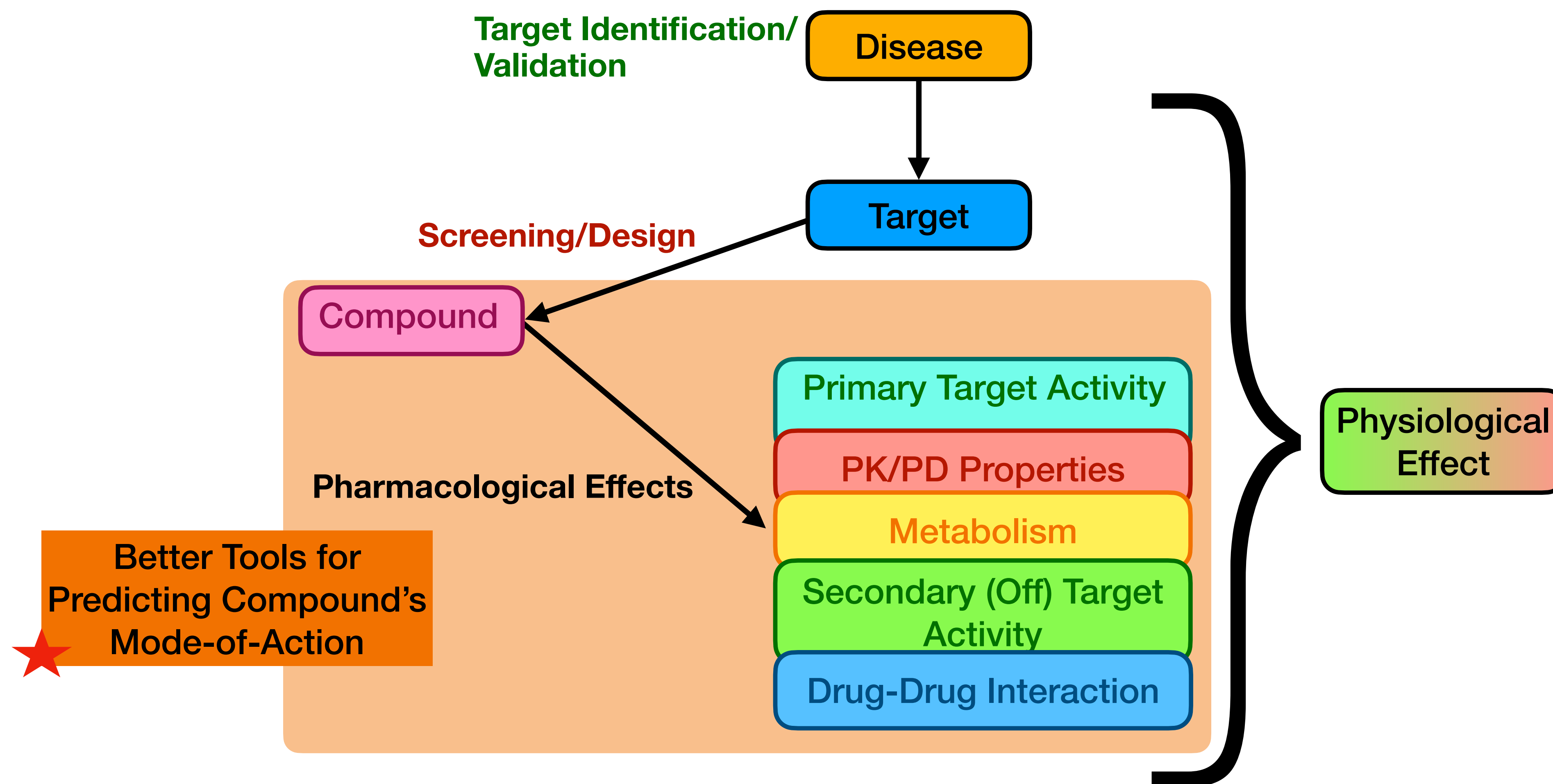


FragmentBERT

Fragment Based Drug Design using Masked Language Models



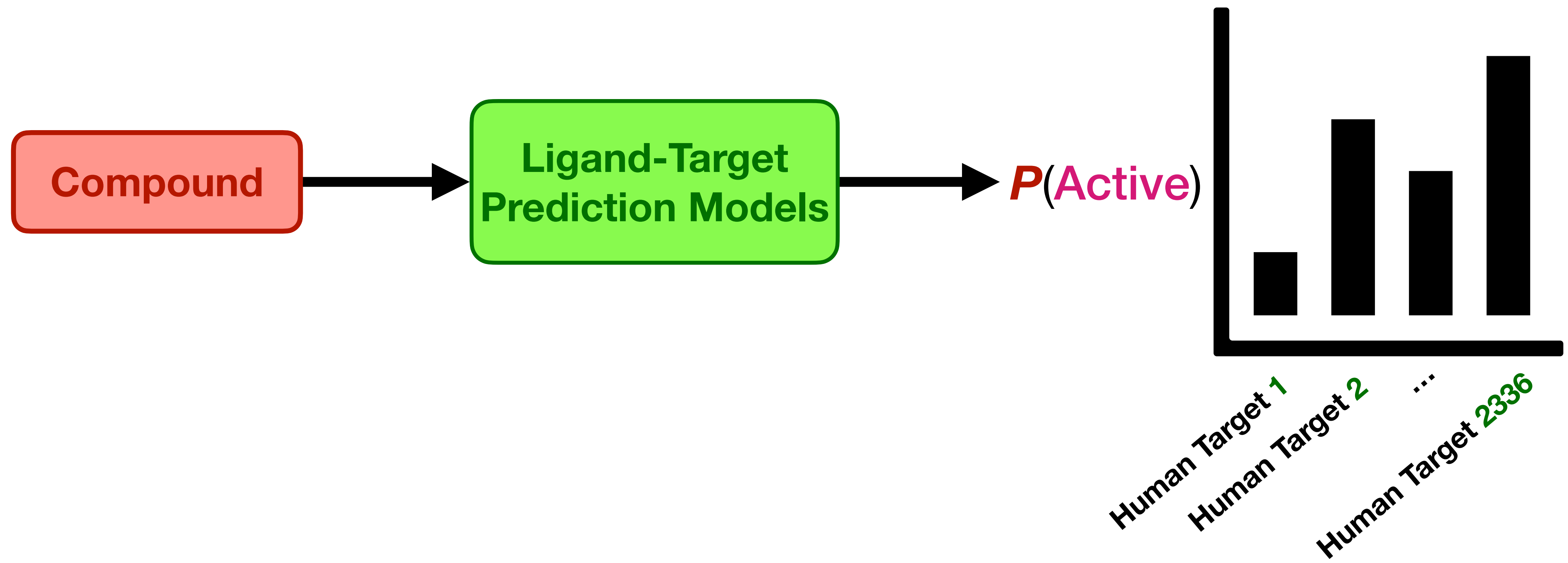
Key Opportunities in Improving *Quality* of Molecules that Enter Clinical Trials



Solution:

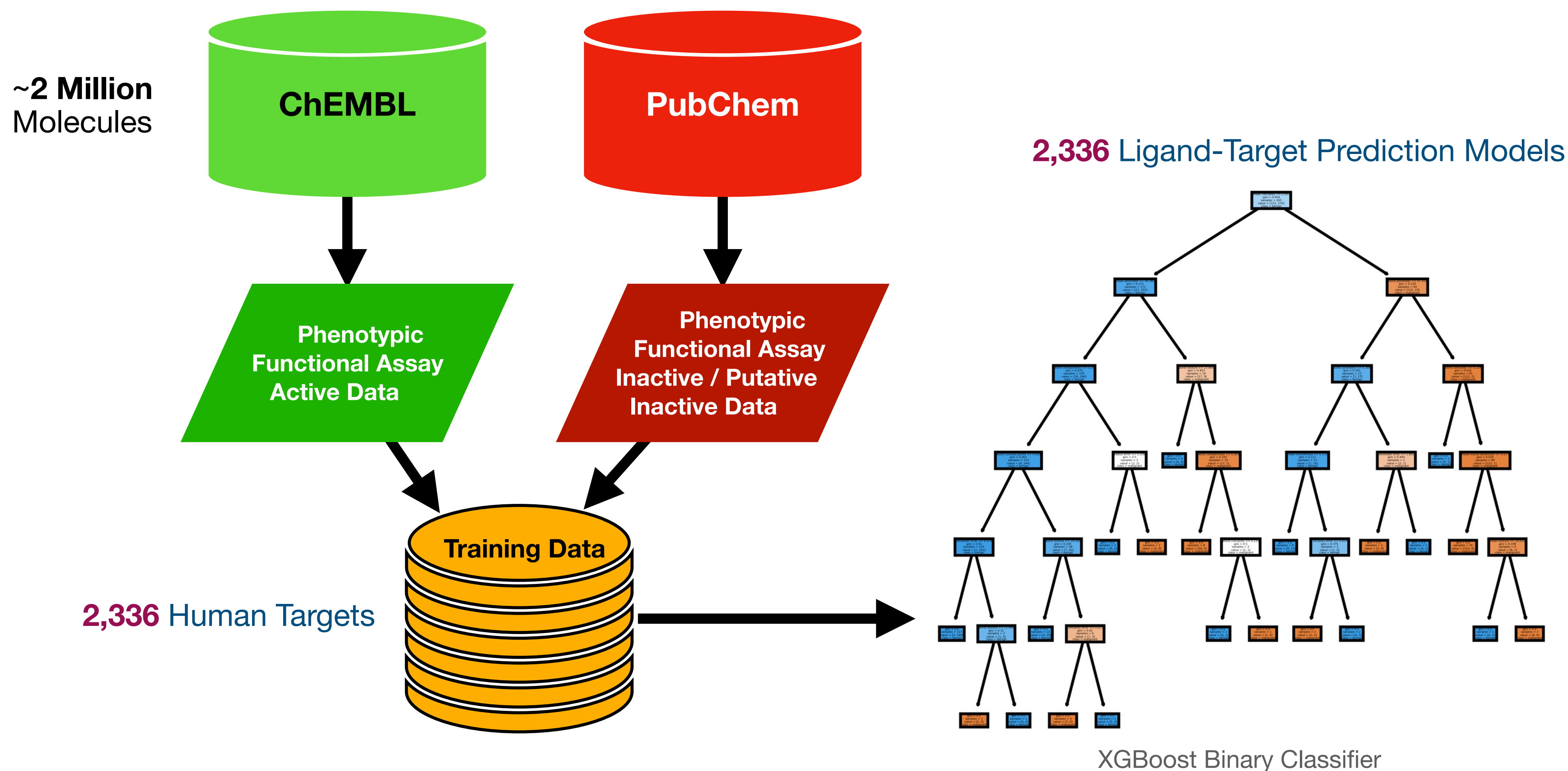
Ligand-Target Prediction
models to predict compound's
Mode-of-Action

Ligand-Target Prediction



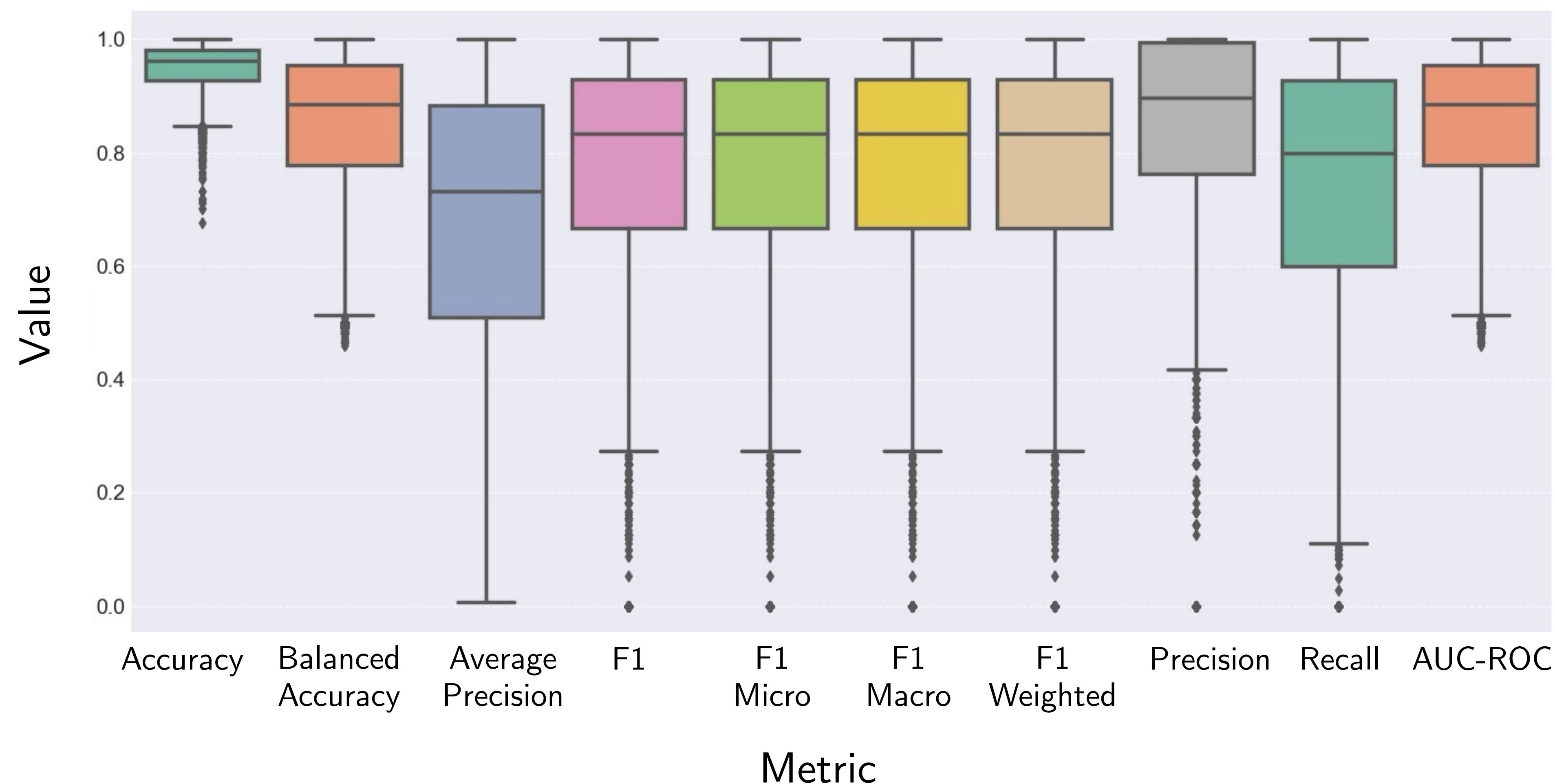
Ligand-Target Prediction

Predicting Functionally Active Targets for Small Molecules



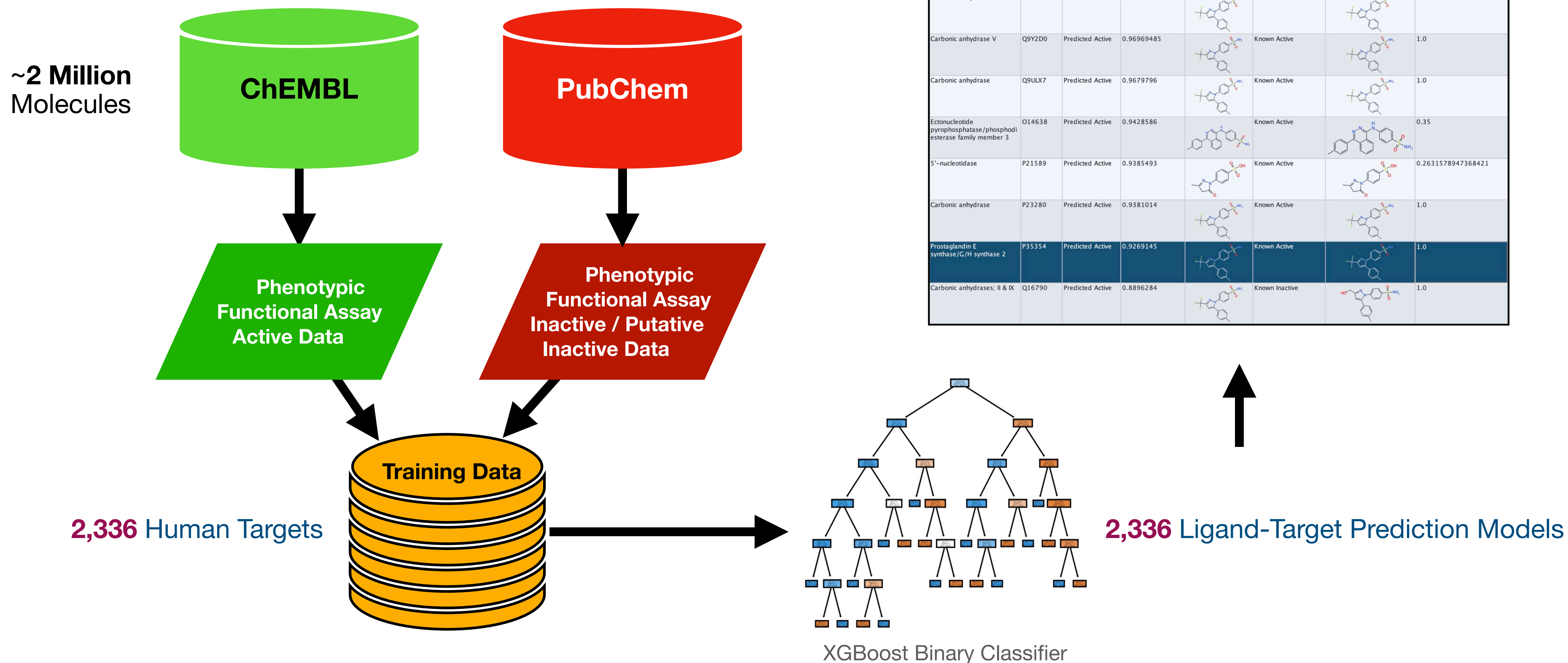
Ligand-Target Prediction

Summary of Model Performance on Holdout Test Set



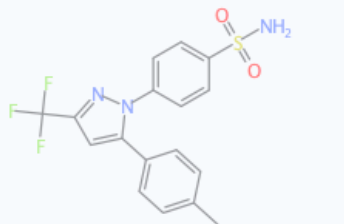
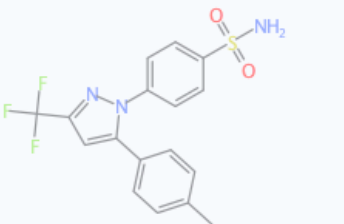
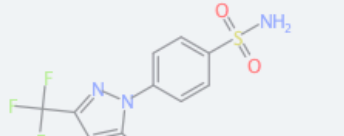
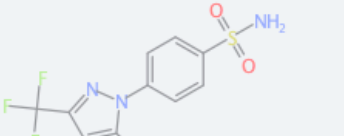
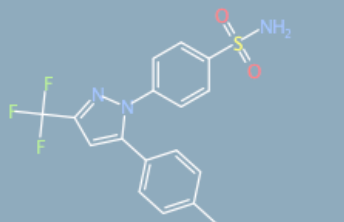
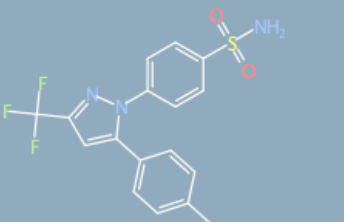
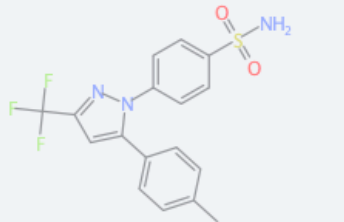
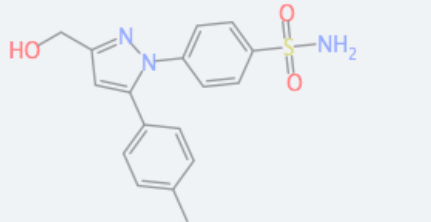
Ligand-Target Prediction

Predicting Functionally Active Targets for Small Molecules



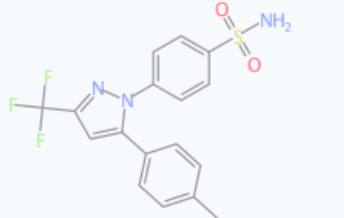
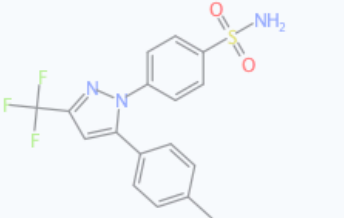
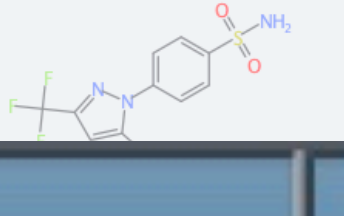
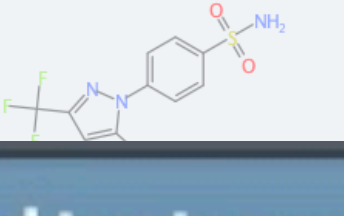
Ligand-Target Prediction

Predicting Functionally Active Targets for Small Molecules

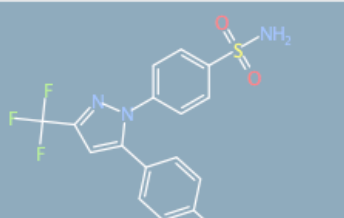
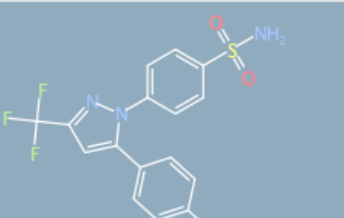
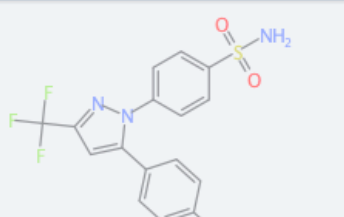
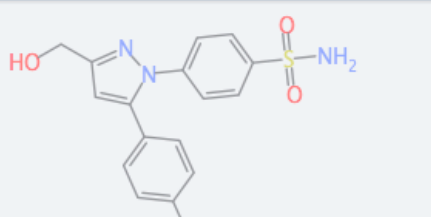
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Carbonic anhydrase V	P35218	Predicted Active	0.97900134		Known Active		1.0
Carbonic anhydrase V	Q9Y2D0	Predicted Active	0.96969485		Known Active		1.0
Carbonic anhydrase	Q16790	Predicted Active	0.8896284		Known Inactive		1.0
Ectonucleotide pyrophosphatase/phosphodiesterase family member 3	O15122	Predicted Active	0.35	Name Uniprot Id Prostaglandin E synthase/G/H synthase 2 P35354			0.2631578947368421
5'-nucleotidase	P28052	Predicted Active	0.9269145				1.0
Carbonic anhydrase	P28052	Predicted Active	0.9269145				1.0
Prostaglandin E synthase/G/H synthase 2	P35354	Predicted Active	0.9269145		Known Active		1.0
Carbonic anhydrases; II & IX	Q16790	Predicted Active	0.8896284		Known Inactive		1.0

Ligand-Target Prediction

Predicting Functionally Active Targets for Small Molecules

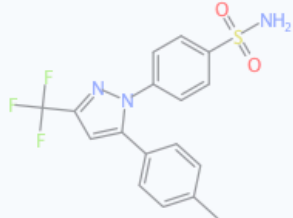
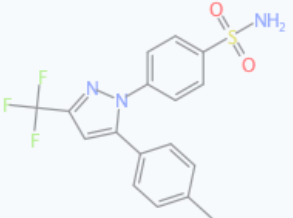
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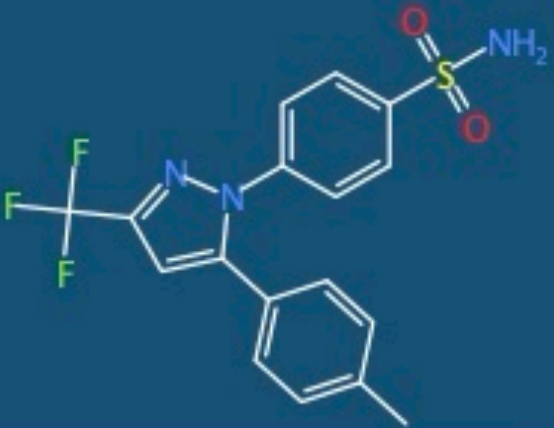
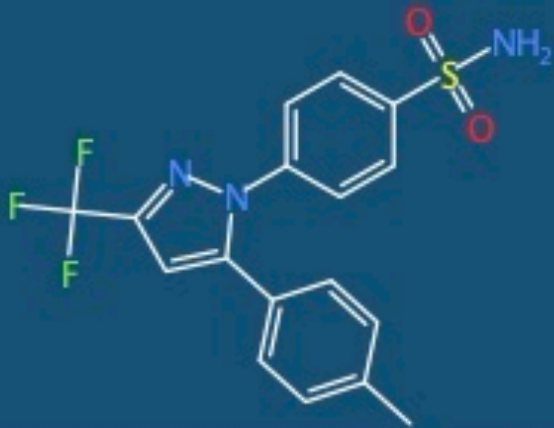
Name	Uniprot Id	Model Prediction @ 10 uM	Prediction probability @ 10 uM
Prostaglandin E synthase/G/H synthase 2	P35354	Predicted Active	0.9269145

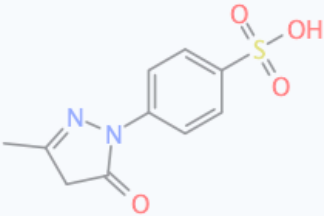
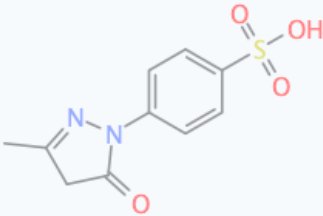
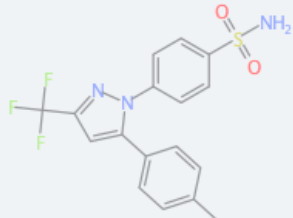
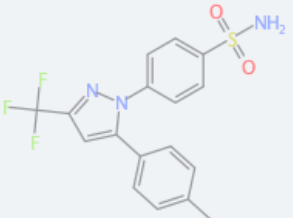
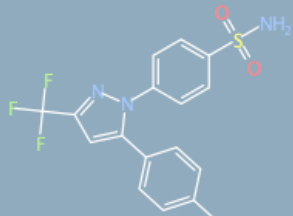
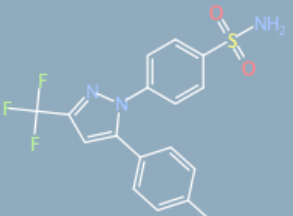
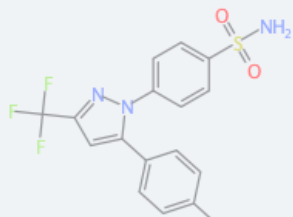
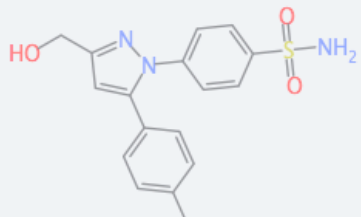
Prostaglandin E synthase/G/H synthase 2	P35354	Predicted Active	0.9269145		Known Active		1.0
Carbonic anhydrases; II & IX	Q16790	Predicted Active	0.8896284		Known Inactive		1.0

Ligand-Target Prediction

Predicting Functionally Active Targets for Small Molecules

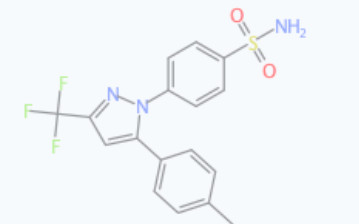
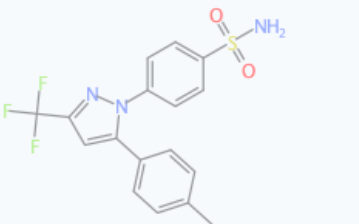
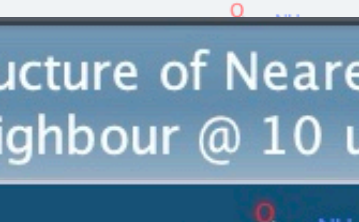
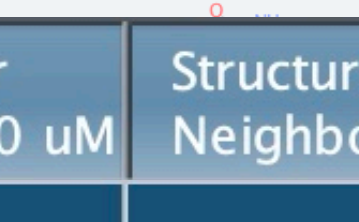
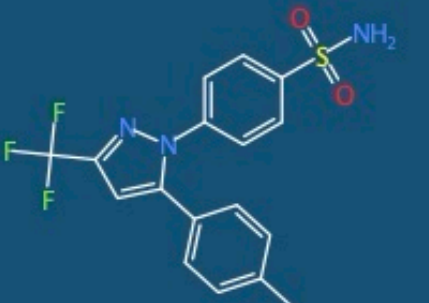
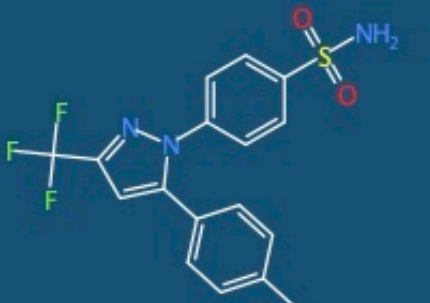
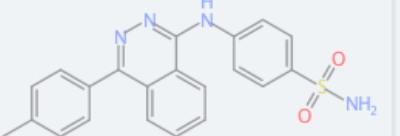
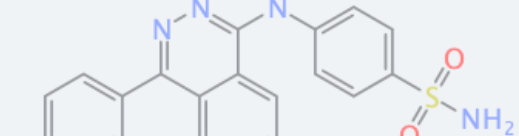
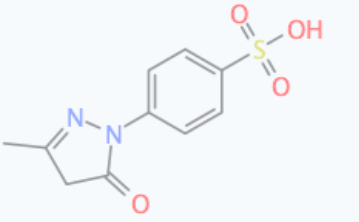
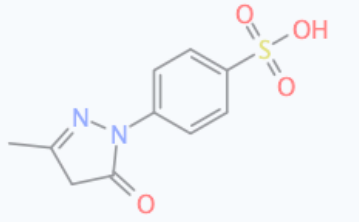
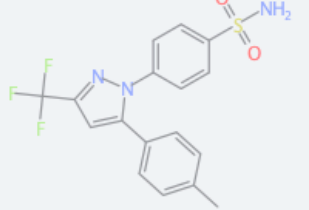
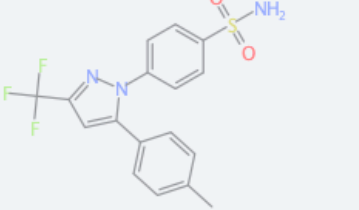
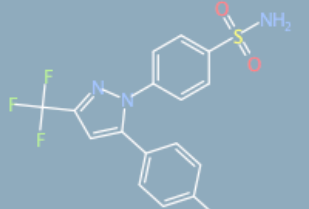
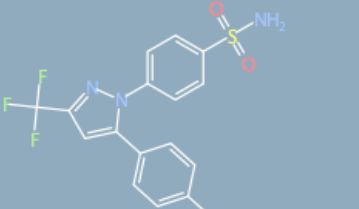
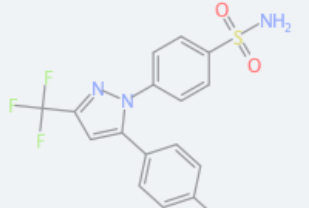
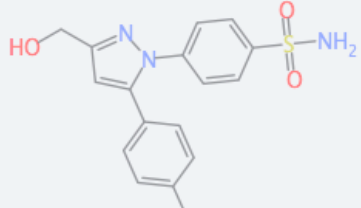
Name	Uniprot Id	Model Prediction @ 10 uM	Prediction probability @ 10 uM	Structure of Nearest Neighbour @ 10 uM	Nearest Neighbour Augmentation @ 10 uM	Structure of Nearest Active Neighbour @ 10 uM	Nearest Neighbour's Tanimoto Similarity @ 10 uM
Carbonic anhydrase V	P35218	Predicted Active	0.97900134		Known Active		1.0

Name	Structure of Nearest Neighbour @ 10 uM	Nearest Neighbour Augmentation @ 10 uM	Structure of Nearest Active Neighbour @ 10 uM	Nearest Neighbour's Tanimoto Similarity @ 10 uM
Prostaglandin E synthase/G/H synthase 2		Known Active		1.0

5'-nucleotidase	P21589	Predicted Active	0.9385493		Known Active		0.2631578947368421
Carbonic anhydrase	P23280	Predicted Active	0.9381014		Known Active		1.0
Prostaglandin E synthase/G/H synthase 2	P35354	Predicted Active	0.9269145		Known Active		1.0
Carbonic anhydrases; II & IX	Q16790	Predicted Active	0.8896284		Known Inactive		1.0

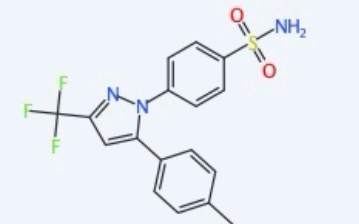
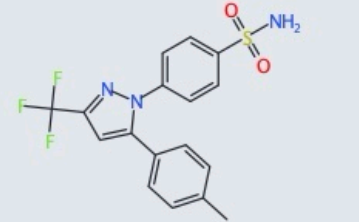
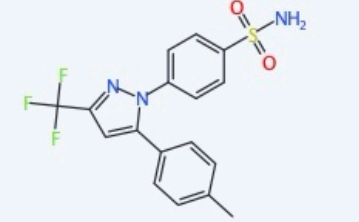
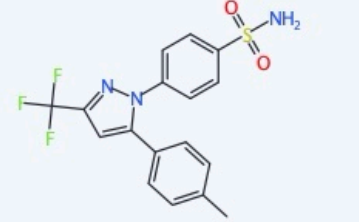
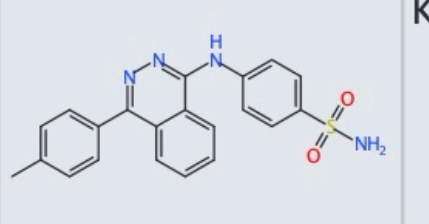
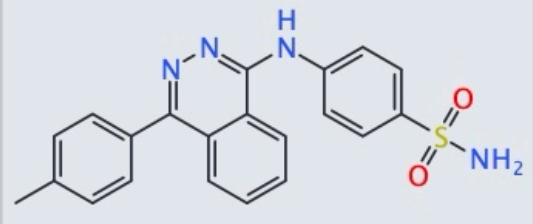
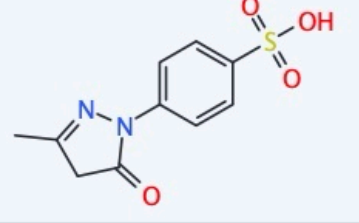
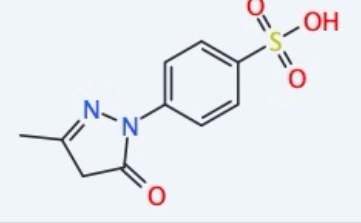
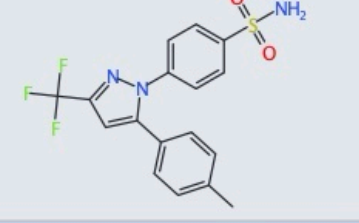
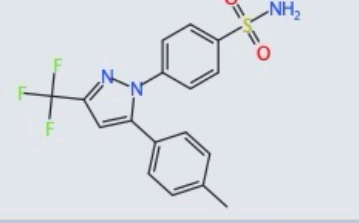
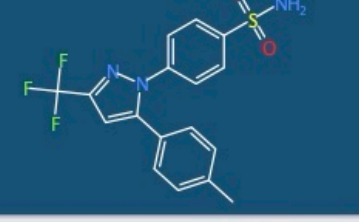
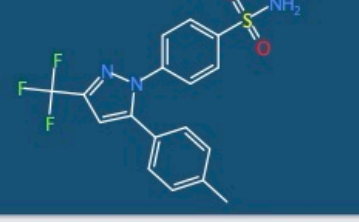
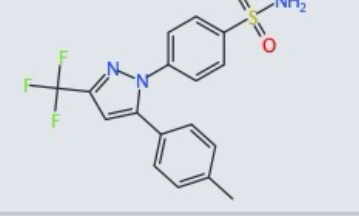
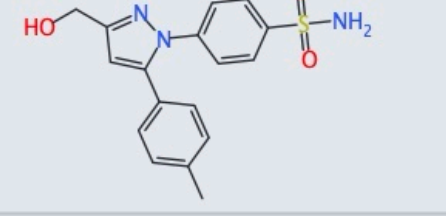
Ligand-Target Prediction

Predicting Functionally Active Targets for Small Molecules

Name	Uniprot Id	Model Prediction @ 10 uM	Prediction probability @ 10 uM	Structure of Nearest Neighbour @ 10 uM	Nearest Neighbour Augmentation @ 10 uM	Structure of Nearest Active Neighbour @ 10 uM	Nearest Neighbour's Tanimoto Similarity @ 10 uM
Carbonic anhydrase V	P35218	Predicted Active	0.97900134		Known Active		1.0
Carbonic anhydrase V	P35218	Predicted Active	0.96969485		Known Active		1.0
Name	Uniprot Id	Model Prediction @ 10 uM	Prediction probability @ 10 uM	Structure of Nearest Neighbour @ 10 uM	Nearest Neighbour Augmentation @ 10 uM	Structure of Nearest Active Neighbour @ 10 uM	Nearest Neighbour's Tanimoto Similarity @ 10 uM
Prostaglandin E synthase/G/H synthase 2	P35354	Predicted Active	0.9269145		Known Active		1.0
pyrophosphatase/phosphodiesterase family member 3							
5'-nucleotidase	P21589	Predicted Active	0.9385493		Known Active		0.2631578947368421
Carbonic anhydrase	P23280	Predicted Active	0.9381014		Known Active		1.0
Prostaglandin E synthase/G/H synthase 2	P35354	Predicted Active	0.9269145		Known Active		1.0
Carbonic anhydrases; II & IX	Q16790	Predicted Active	0.8896284		Known Inactive		1.0

Ligand-Target Prediction

Predicting Functionally Active Targets for Small Molecules

Name	Uniprot Id	Model Prediction @ 10 uM	Prediction probability @ 10 uM	Structure of Nearest Neighbour @ 10 uM	Nearest Neighbour Augmentation @ 10 uM	Structure of Nearest Active Neighbour @ 10 uM	Nearest Neighbour's Tanimoto Similarity @ 10 uM
Carbonic anhydrase V	P35218	Predicted Active	0.97900134		Known Active		1.0
Carbonic anhydrase V	Q9Y2D0	Predicted Active	0.96969485		Known Active		1.0
Carbonic anhydrase	Q9ULX7	Predicted Active	0.9679796		Known Active		1.0
Ectonucleotide pyrophosphatase/phosphodiesterase family member 3	O14638	Predicted Active	0.9428586		Known Active		0.35
5'-nucleotidase	P21589	Predicted Active	0.9385493		Known Active		0.2631578947368421
Carbonic anhydrase	P23280	Predicted Active	0.9381014		Known Active		1.0
Prostaglandin E synthase/G/H synthase 2	P35354	Predicted Active	0.9269145		Known Active		1.0
Carbonic anhydrases; II & IX	Q16790	Predicted Active	0.8896284		Known Inactive		1.0

& 2328 more...

Ligand-Target Prediction

Mode-of-Action Analysis of Z-Guggulsterone for Triple Negative Breast Cancer

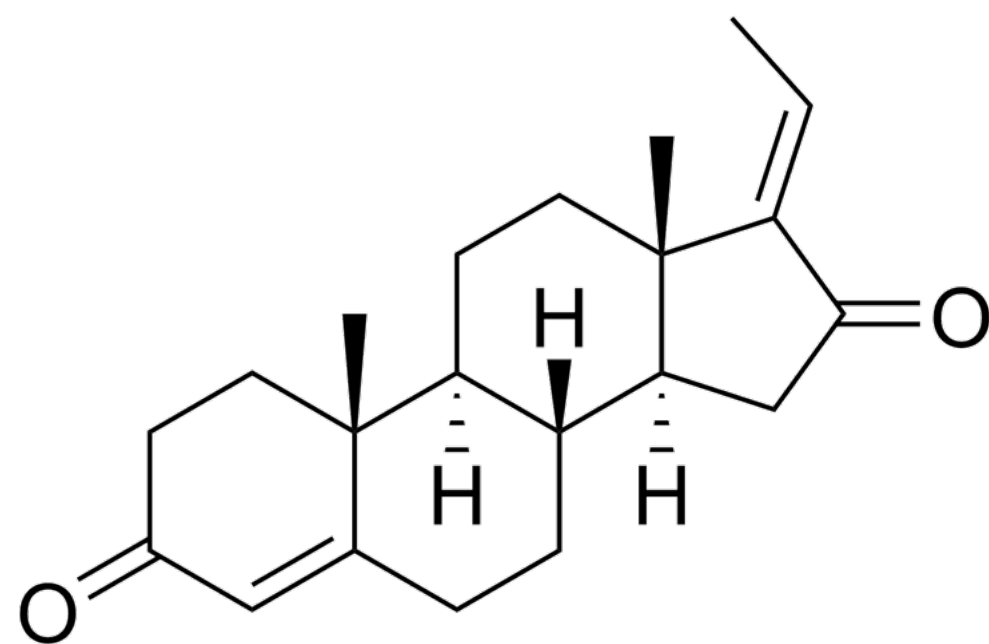


Environmental Research
Volume 232, 1 September 2023, 116335

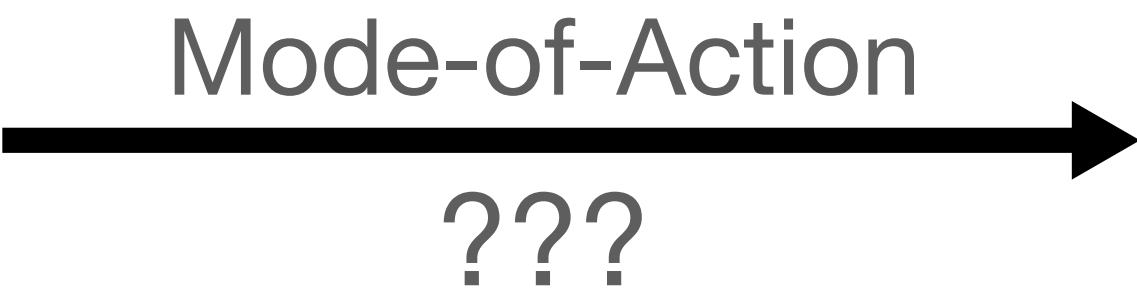


CXCR4 expression is elevated in TNBC patient derived samples and Z-guggulsterone abrogates tumor progression by targeting CXCL12/CXCR4 signaling axis in preclinical breast cancer model

IF = 8.3

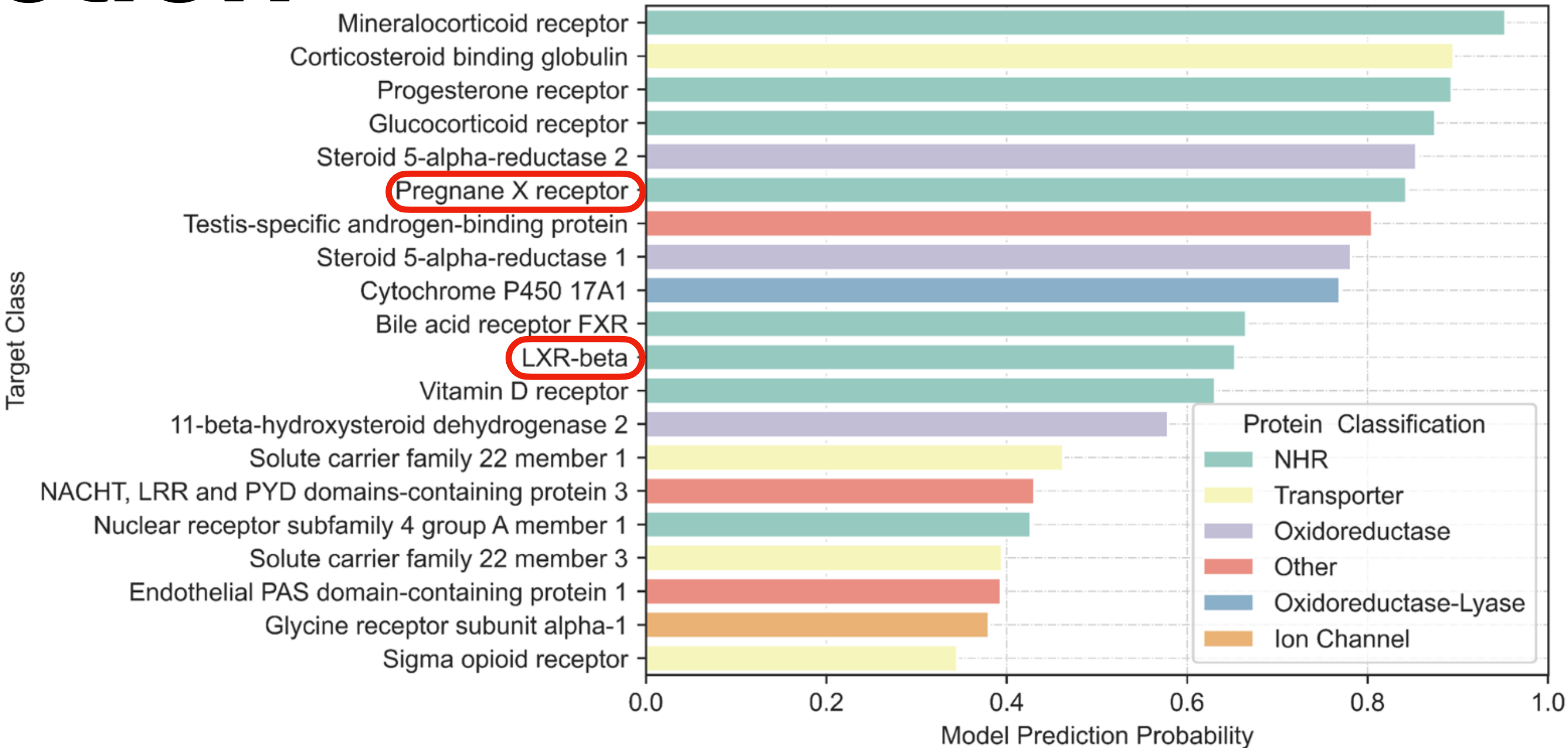
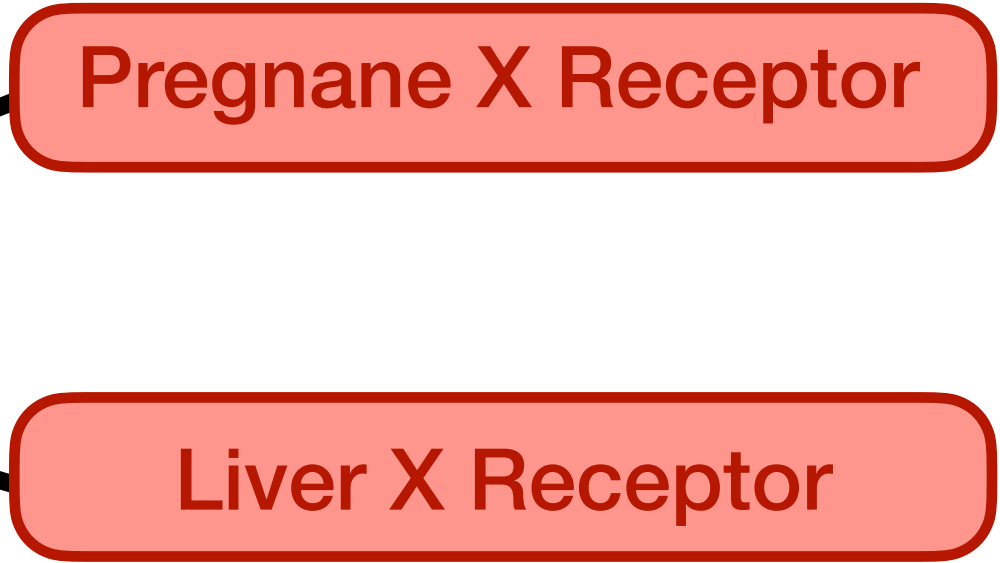


Z-Guggulsterone



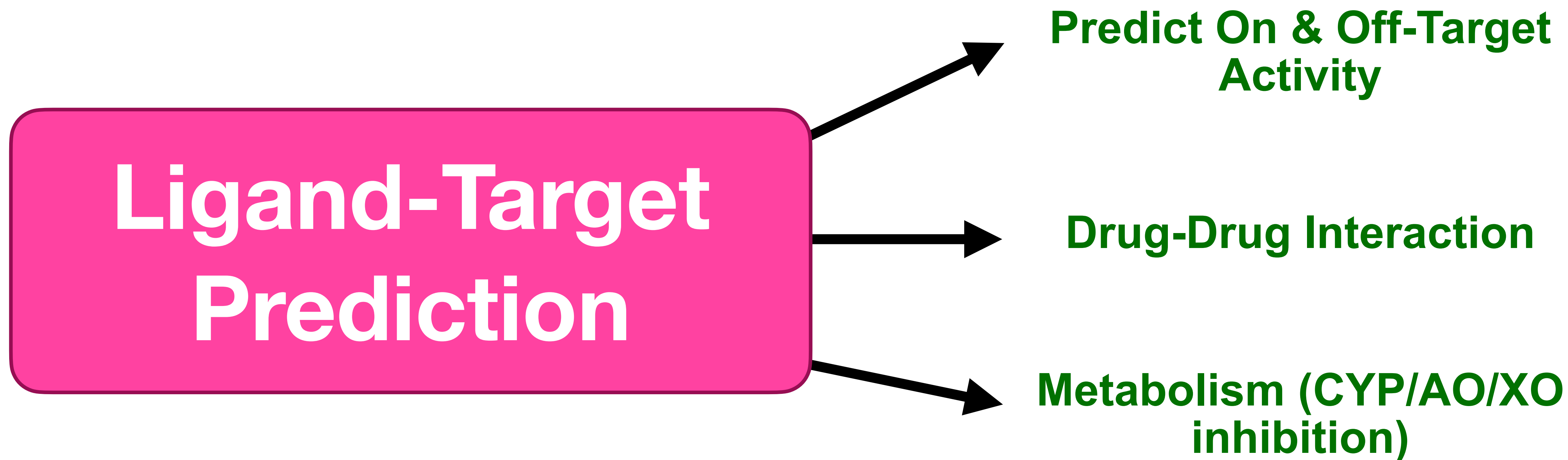
Triple Negative Breast Cancer

Ligand-Target Prediction Model Probabilities



Ligand-Target Prediction

Molecular Mode-of-Action Analysis Prediction



FragmentBERT + Ligand-Target Prediction

Optimizing Molecular Generation using Mode-of-Action Analysis

FragmentBERT Code is
Open Sourced under
Apache License v2



Hardware
Accelerated
(GPU / TPU)
Inference

FragmentBERT

- Fragment Linking
- Fragment Merging
- Fragment Growth
- Synthetically Accessible Library Generation

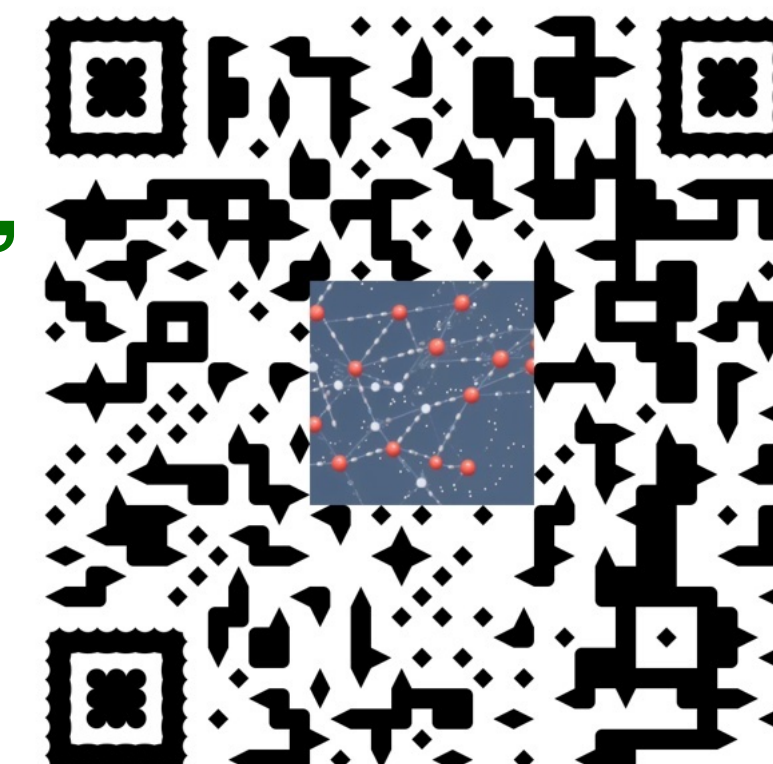
Predict Functional
Activity on Targets
of Pharmacological
Importance

Lower Compute,
Faster Outputs

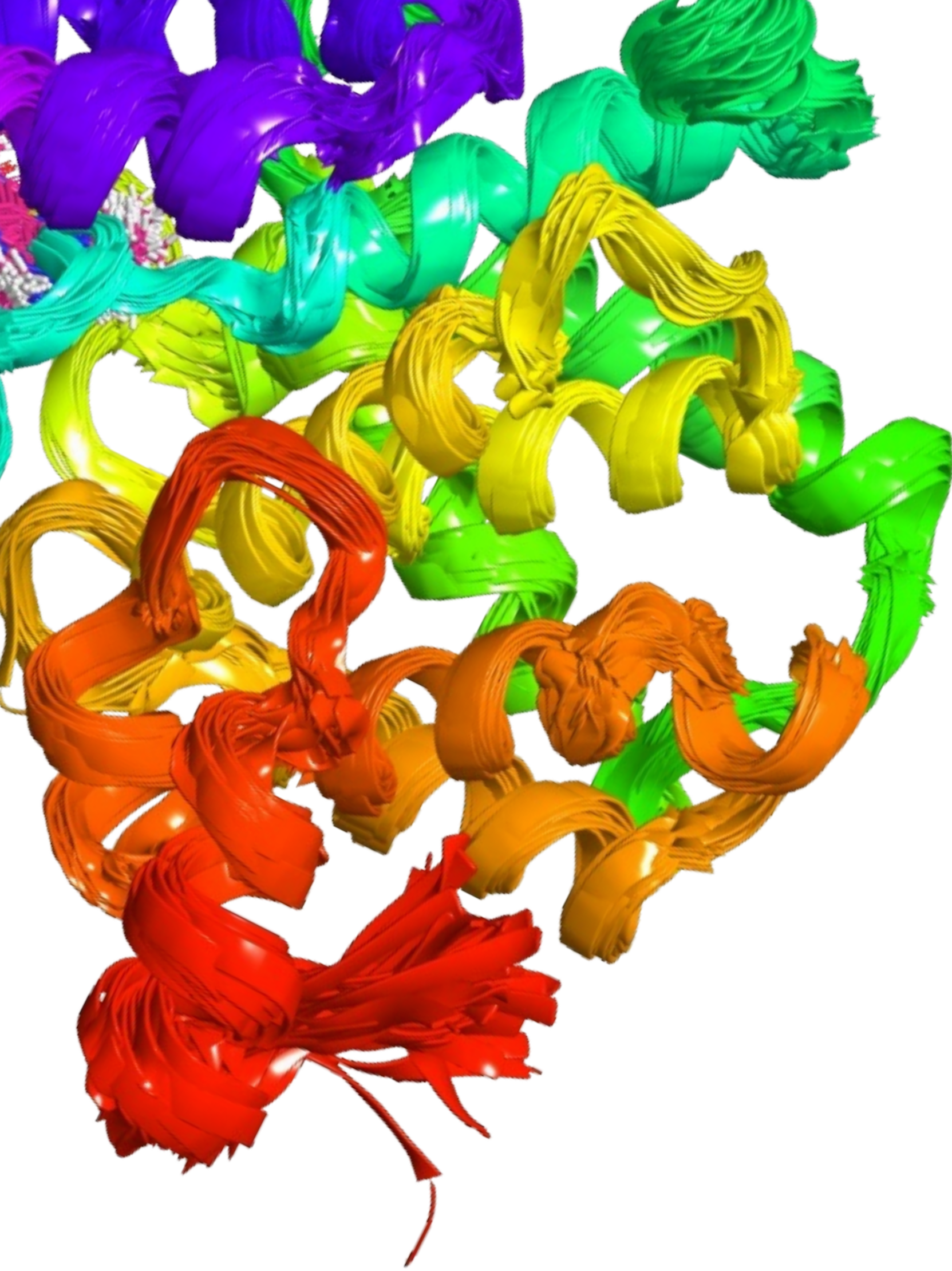
Ligand-Target Prediction

Fragment Based
Drug-Like Ligand
Generation

Ligand-Target
Prediction Models are
Open Sourced under
Apache License v2



- On & Off (secondary) Target Activity
- Metabolism (CYP/AO/XO inhibition)
- Drug-Drug Interaction



Thank You !!

